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Abstract

Transformer representations admit reliable linear access through probes, sparse autoencoders, and activation steering, despite the depth and nonlinearity of the underlying networks. We give an architectural-statistical account of why. Architecturally, linear unembedding makes every logit margin between two output tokens a projection of the residual stream onto a fixed contrast direction, and the cross-entropy gradient pushes representations along weighted combinations of the same directions; under nonlinear readouts, this necessity disappears. Statistically, under sparse low-coherence superposition, projection onto fixed feature directions recovers feature magnitudes up to an explicit interference bound. A direct consequence is the *Self-Reference Property*: class prompts are themselves contexts that express a feature, so their activations provide approximate references for that feature direction without supervision. We validate the picture across eight classification tasks, four model families, and three architectures (decoder-only LLMs, BERT, ViT/LLaVA), with extensions to cross-modal, compositional tasks and attribution patching. Linear interpretability emerges as a predictable consequence of how transformers compute—fixed read and write directions modulated by context-dependent magnitudes—rather than an unexplained empirical regularity.

1 Introduction

Linear structure has become a central organizing principle in modern interpretability for transformers [Vaswani et al., 2017]. Linear probes recover semantic attributes from hidden states [Alain and Bengio, 2016, Belinkov, 2022], sparse autoencoders (SAE) identify interpretable feature directions [Bricken et al., 2023, Cunningham et al., 2023], and single-vector activation steering reliably modifies model behavior [Turner et al., 2023, Zou et al., 2023]. Across a wide range of methods and objectives, simple linear operations repeatedly succeed at isolating meaningful internal structures.

Yet this success raises a fundamental theoretical puzzle. Transformers are deep, highly nonlinear systems trained with simple objectives but exhibiting complex emergent behavior. Why, then, should their internal representations admit such simple and reliable linear access to semantic information?

We propose that transformer representations are organized as a system of fixed feature directions whose magnitudes vary with context. Each semantic feature corresponds to a stable direction in residual space; the activation for any context decomposes as a sparse sum over these directions; and across all contexts expressing a given feature, the direction stays the same—only the scalar magnitude varies. Linear interpretability methods recover meaningful structure because they read the same directional quantities the architecture itself reads. This picture is not a claim about *why* features emerge—emergence depends on optimization dynamics, training data, and inductive biases that existing work studies. Our claim concerns the *form* features take whenever they emerge and are reused: any feature consumed through the model’s fixed linear interfaces is exposed through a fixed direction, regardless of how it came to be there.

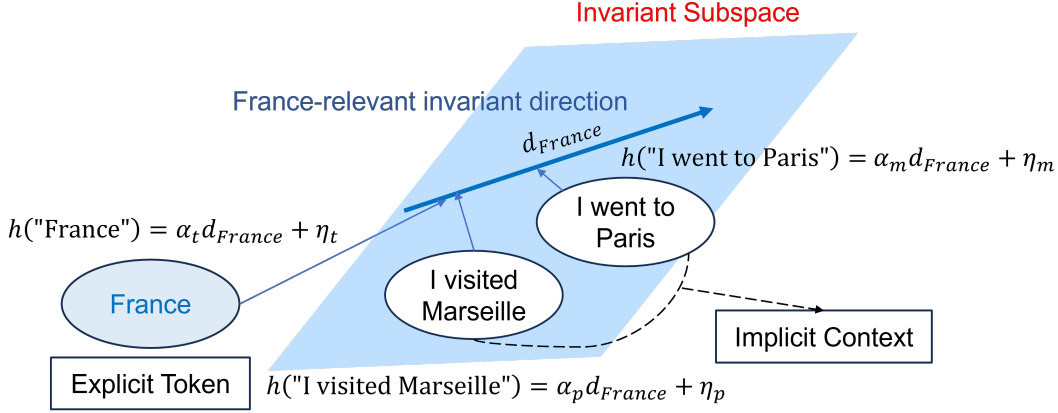


Figure 1: **Context-invariant directional representation.** The explicit token “France” provides a reference vector for this direction (*self-reference*), while contextual mentions such as “I went to Paris” and “I visited Marseille” share the same invariant direction.

This picture has both architectural and statistical content. Architecturally, the linearity of the unembedding makes every logit margin between two output tokens a projection of the residual stream onto a fixed contrast direction, and the cross-entropy gradient pushes representations along weighted combinations of the same directions (Proposition 3.1); the same fixed-coordinate structure is not architecturally required under nonlinear readouts (Proposition 3.2), so linearity is not a consequence of cross-entropy alone but of cross-entropy applied through linear or selection-only readout interfaces. Statistically, sparse low-coherence superposition makes feature magnitudes recoverable from residual activations by projection up to an explicit interference bound (Theorem 3.3); when unembedding rows decompose as feature bundles, next-token prediction reduces to weighted bundle matching (Proposition A.1). The framework also extends to features written by 2-layer MLP blocks under a *selection-only* decomposition; deeper MLPs preserve write-direction structure but couple feature magnitudes through compositional gating.

A direct consequence is the *Self-Reference Property*: a class prompt is itself a context where the corresponding feature is active, so its activation provides an approximate reference for that feature’s direction in a single forward pass, with no labeled instance data or learned probes (Corollary A.29). The activation produced by “France” is approximately aligned with the activation produced by “the country of the Eiffel Tower”; either can serve as a reference for locating the feature in other representations.

We validate the picture across eight semantic classification tasks and four model families (LLaMA3-8B [Grattafiori et al., 2024], Mistral-7B [Jiang et al., 2023], GPT2-Small [Radford et al., 2019], LLaMA3.2-3B [Meta AI, 2024]), as well as BERT, ViT, and LLaVA on cross-architecture and cross-modal tasks.

In summary, our contributions are: (1) an architectural-statistical account of why transformer representations are organized as fixed feature directions modulated by context-dependent magnitudes, with the readout’s role isolated by the contrast between linear and nonlinear unembeddings; (2) the *Self-Reference Property*, which establishes that class prompts provide approximate empirical references for feature directions and yields a zero-shot probe requiring no labeled instance data; (3) convergent empirical evidence across eight tasks, four model families, and three architectures, including SAE-recovered features that align with class-token directions.

2 Related Work

2.1 Mechanistic Interpretability

Mechanistic interpretability seeks to explain model behavior by identifying internal features, circuits, and causal mechanisms inside neural networks rather than relying solely on input–output correlations [Elhage et al., 2021, Nanda et al., 2023]. Transformer models contain identifiable computational

substructures implementing specific behaviors, and targeted interventions on internal activations can causally alter model outputs [Meng et al., 2022, Wang et al., 2022, Turner et al., 2023, Zou et al., 2023]. Our work connects this literature to representation geometry by providing a theoretical account of the linear, context-invariant feature structure that underlies many of these observations.

2.2 Linear Representation

Semantic attributes can often be captured by low-dimensional linear structure in representation spaces, dating back to linear regularities in word embeddings [Mikolov et al., 2013, Pennington et al., 2014]. In modern LMs, linear structure is operationalized as *measurement* (linear probes that extract a property) and as *intervention* (steering representations along a direction) [Kim et al., 2018, Ravfogel et al., 2020]. Park et al. [2024] formalize these intuitions, clarifying what “linear representation” means and how measurement and intervention relate geometrically.

Relation to optimization-based accounts. Jiang et al. [2024] show that next-token prediction with softmax cross-entropy and the implicit bias of gradient descent can promote linear concept representations. Their setting assumes a linear unembedding; we make this dependence load-bearing. Proposition 3.2 shows that without a linear readout, the same task admits solutions whose target features are not linearly decodable, and the grokking ablation (Section 5.1) confirms the distinction empirically. Optimization supplies the pressure to learn task-relevant features; the readout architecture determines the coordinates through which that pressure operates.

Related invariant-subspace perspectives. Two recent lines of work converge on related geometric pictures. Schiffman [2026] extract *algorithmic cores*—compact subspaces necessary and sufficient for task performance—and show they recur across independently trained transformers and across scales, supporting the view that transformer computation is organized around low-dimensional invariants. Ahmad et al. [2025] decompose attention heads and MLPs into orthogonal singular directions and find that previously identified canonical functional heads encode multiple overlapping subfunctions aligned with distinct singular directions. Our framework is complementary: where these works empirically identify low-dimensional structure that recurs across runs or within components, we provide an architectural and statistical account of *why* such structure must take a fixed-direction form whenever it is consumed by linear interfaces.

2.3 Linear Probes

Linear probes test what information is present in hidden states [Alain and Bengio, 2016, Hewitt and Manning, 2019, Tenney et al., 2019]. A range of successful techniques instantiate this idea: the *logit lens* decodes intermediate representations using the unembedding matrix, revealing that hidden states already resemble next-token distributions [nostalgebraist, 2020], and the tuned lens extends this with layer-specific affine transformations [Belrose et al., 2023]. Relatedly, linear directions identified by probes have been used not only for measurement but also for intervention, enabling activation steering of hidden states [Li et al., 2023, Rinsky et al., 2024]. Recent evaluations further show that sparse autoencoder latents do not consistently outperform simple linear baselines on probing tasks [Kantamneni et al., 2025]. Together, these results highlight the surprising effectiveness of linear readouts and interventions across diverse settings. However, they remain largely descriptive and do not explain *why* such linear decodability should arise in deep nonlinear models. Our theory addresses this gap.

3 Theoretical Framework

We propose that transformer representations are organized as a system of fixed feature directions modulated by context-dependent magnitudes. Concretely: each semantic feature f corresponds to a stable direction $\mathbf{d}_f$ in residual space; the activation $\mathbf{h}(c)$ at the position of interest in context c decomposes as a sparse sum $\sum_{g \in S(c)} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c)$; and across all contexts expressing f , the direction $\mathbf{d}_f$ is the same—only the magnitude $\alpha_f(c)$ varies. The transformer’s read and write operations are linear maps acting on this representation: attention QKV projections and the unembedding read scalar quantities along fixed directions; attention OV outputs and 2-layer MLP write-back paths add fixed columns scaled by context-dependent gates. Linear interpretability succeeds because it reads

the same scalar magnitudes the architecture itself reads. This is a claim about the geometric *form* that reused features take when they are exposed through the model’s fixed linear interfaces, not a claim about which features emerge during training; emergence is a separate question addressed by work on optimization and inductive biases.

Setup. We consider standard decoder-only transformers with an additive residual stream, fixed linear interfaces (attention W_Q, W_K, W_V, W_O , and unembedding W_U), and a linear output layer $\ell(c) = W_U \mathbf{h}^{(L)}(c)$. We write $\mathbf{u}_t = (W_U)_{t,:}$ for the unembedding row of token t , and $\mathbf{h}(c) := \mathbf{h}^{(L)}(c)$ when no ambiguity arises. Throughout, we adopt the sparse-superposition representation model from prior work [Elhage et al., 2022]: residual activations decompose over a unit-norm dictionary $\mathcal{D} = \{\mathbf{d}_g\}_{g \in \mathcal{F}}$ with coherence $\mu = \max_{g \neq g'} |\langle \mathbf{d}_g, \mathbf{d}_{g'} \rangle|$, sparse support $|S(c)| \leq k$, and bounded residual $\|\xi(c)\| \leq \varepsilon$. We further assume that unembedding rows admit an analogous bundle decomposition $\mathbf{u}_t = \sum_{f \in B(t)} b_{t,f} \mathbf{d}_f + \mathbf{r}_t$ with $\|\mathbf{r}_t\| \leq \rho_t$, and that the architectural readout direction $\mathbf{w}_f = L_f^\top \phi_f$ aligns with the dictionary direction up to a misalignment parameter γ , $\langle \mathbf{w}_f, \mathbf{d}_f \rangle / \|\mathbf{w}_f\| \geq 1 - \gamma$. Full statements, motivation, and all proofs appear in Appendix A.

3.1 Architectural content: fixed contrast coordinates

The first content of the framework is that linear unembedding makes both inference and training operate through the same set of fixed coordinates.

Proposition 3.1 (Linear Unembedding Imposes Fixed Contrast Coordinates). *With logits $\ell(c) = W_U \mathbf{h}(c)$, the logit margin between any two output tokens is $\ell_y(c) - \ell_j(c) = (\mathbf{u}_y - \mathbf{u}_j)^\top \mathbf{h}(c)$, the projection of the residual stream onto the fixed contrast direction $\mathbf{v}_{y,j} := \mathbf{u}_y - \mathbf{u}_j$. For cross-entropy training with target token y and softmax probabilities $\mathbf{p}(c) = \text{softmax}(\ell(c))$, the gradient at the final residual stream is*

$$\nabla_{\mathbf{h}} \mathcal{L}(c) = W_U^\top (\mathbf{p}(c) - \mathbf{e}_y) = \sum_{j \neq y} p_j(c) (\mathbf{u}_j - \mathbf{u}_y),$$

a probability-weighted sum of the same fixed contrast directions that govern inference.

The architectural premise here is the linearity of the unembedding—a fact about standard transformer design, not an additional assumption about features. The conclusion is that the same fixed contrast directions $\mathbf{u}_y - \mathbf{u}_j$ determine both how the model decides between two tokens and how training updates push the residual stream during learning. Cross-entropy is not arbitrary pressure: it rewards making evidence for the correct token accessible through the same coordinates the model uses to compare tokens. The proposition does not claim that semantic features must emerge; it states that *if* a feature ends up affecting the model’s choice between two tokens, it must do so through the corresponding fixed contrast coordinate. Whether such features emerge in the first place depends on optimization dynamics, training data, and inductive biases, which existing work studies; the architectural account here characterizes the geometric form in which any such feature is exposed.

The fixed-coordinate property of Proposition 3.1 is architecturally necessary under linear readouts. The following counterfactual proposition isolates the role of the readout: under nonlinear readouts, the same fixed-coordinate structure is no longer required.

Proposition 3.2 (Nonlinear Readouts Remove Fixed-Coordinate Necessity). *Let logits be produced by a nonlinear readout $\ell = G(\mathbf{h})$, and define the margin between output tokens y and j as $m_{y,j}(\mathbf{h}) = G_y(\mathbf{h}) - G_j(\mathbf{h})$. Then*

$$\nabla_{\mathbf{h}} m_{y,j}(\mathbf{h}) = \nabla_{\mathbf{h}} G_y(\mathbf{h}) - \nabla_{\mathbf{h}} G_j(\mathbf{h}),$$

which depends on $\mathbf{h}$ in general. Unless this gradient is constant over the relevant domain of $\mathbf{h}$, no fixed direction $\mathbf{v}_{y,j}$ and scalar b exist with $m_{y,j}(\mathbf{h}) = \mathbf{v}_{y,j}^\top \mathbf{h} + b$. Nonlinear readouts therefore remove the architectural necessity that output-relevant evidence be exposed through a single fixed contrast coordinate; the same task can in principle be solved by representations whose target features are not linearly decodable.

Proof. A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is affine ($f(\mathbf{h}) = \mathbf{v}^\top \mathbf{h} + b$ for fixed $\mathbf{v}, b$) if and only if $\nabla_{\mathbf{h}} f$ is constant. The margin gradient $\nabla_{\mathbf{h}} m_{y,j} = \nabla_{\mathbf{h}} G_y - \nabla_{\mathbf{h}} G_j$ is constant in $\mathbf{h}$ if and only if both G_y and G_j are affine up to a common nonlinear part that cancels in the difference. For generic nonlinear G (e.g., a deep MLP), this fails, and $m_{y,j}$ is not affine. $\square$

Together, Propositions 3.1 and 3.2 isolate the architectural contribution: linear readouts make fixed contrast coordinates a necessary form for output-relevant evidence; nonlinear readouts do not (Example 1 in Appendix A gives a concrete XOR-style instance). Prior work establishing that linear concept representations emerge under cross-entropy training [Jiang et al., 2024] is conducted in models with linear unembeddings, where the linearity of the readout is treated as a fixed background condition. Proposition 3.2 makes this conditioning visible: the linearity of features observed in standard transformers is not a consequence of cross-entropy and gradient descent in isolation, but of cross-entropy and gradient descent applied through linear or selection-only readout interfaces. The grokking ablation in Section 5.1 provides direct empirical evidence: under identical task, optimizer, and training dynamics, replacing the linear unembedding with a deep MLP head disrupts the consistent emergence of Fourier directional structure across seeds.

The training-time picture mirrors the inference-time one. For a general nonlinear readout, the gradient at the final residual stream becomes $J_G(\mathbf{h})^\top (\mathbf{p}(c) - \mathbf{e}_y)$, with rows of J_G that depend on context. A 2-layer MLP readout $G(\mathbf{h}) = W_{\text{out}}\sigma(W_{\text{in}}\mathbf{h} + \mathbf{b})$ preserves a selection-only version: the gradient $\nabla_{\mathbf{h}}\mathcal{L} = W_{\text{in}}^\top \text{diag}(\sigma'(\cdot))W_{\text{out}}^\top (\mathbf{p}(c) - \mathbf{e}_y)$ remains a sum over the fixed columns of $W_{\text{in}}^\top$, with only the scalar gates $\sigma'(\cdot)$ depending on c . Deeper readouts compose multiple gates and break this independent-gating structure (Proposition A.1, Appendix A). The same interface-local picture applies to attention: the QKV projections W_Q, W_K, W_V are themselves fixed linear interfaces that read the residual stream along fixed directions, while the input-dependent softmax determines *how much* content is routed across positions but not *which* directions are read; consequently, features consumed by an attention head are exposed at that head’s input through the same fixed-direction structure that governs the unembedding, only locally to that head rather than globally.

Why shared directions are an efficient solution. Propositions 3.1 and 3.2 characterize what is and is not architecturally required, but not why training tends to produce reusable shared directions. We show in Appendix A (Proposition A.10) that under a linear readout, the minimum-norm representation achieving a required margin on the (y, j) contrast is the line spanned by $\mathbf{u}_y - \mathbf{u}_j$, with magnitude scaled by the required margin: contexts requiring the same contrast collapse onto a shared direction with context-dependent scalar. The same minimum-norm problem under a nonlinear margin admits no such collapse in general (Remark A.11). Standard mechanisms in real transformers produce a related efficiency bias—weight decay, the implicit bias of gradient descent toward small-norm solutions, and LayerNorm’s bounded activation norms—so the proposition characterizes the geometric form that efficient solutions take when found, rather than deriving feature emergence.

3.2 Statistical content: projection error and its decomposition

Architectural pressure on directions is necessary but not sufficient for projection-based interpretability to work: the recovered scalar must actually approximate the underlying feature magnitude. We characterize the error of projection-based recovery exactly, identifying two interference terms that govern when the bound is tight.

Theorem 3.3 (Projection Error Decomposition). *Let $\mathbf{h}(c) = \alpha_f(c)\mathbf{d}_f + \sum_{g \in S(c), g \neq f} \alpha_g(c)\mathbf{d}_g + \boldsymbol{\xi}(c)$ with $\|\mathbf{d}_f\| = 1$. Then projection onto $\mathbf{d}_f$ decomposes exactly as*

$$\langle \mathbf{h}(c), \mathbf{d}_f \rangle = \alpha_f(c) + I_f(c) + R_f(c),$$

where

$$I_f(c) := \sum_{g \in S(c), g \neq f} \alpha_g(c) \langle \mathbf{d}_g, \mathbf{d}_f \rangle$$

is the cross-feature interference and $R_f(c) := \langle \boldsymbol{\xi}(c), \mathbf{d}_f \rangle$ is the residual contamination. Therefore

$$|\langle \mathbf{h}(c), \mathbf{d}_f \rangle - \alpha_f(c)| \leq |I_f(c)| + |R_f(c)|.$$

If additionally $|\langle \mathbf{d}_g, \mathbf{d}_f \rangle| \leq \mu$ for $g \neq f$, $|S(c)| \leq k$, $|\alpha_g(c)| \leq A$, and $\|\boldsymbol{\xi}(c)\| \leq \varepsilon$, then the worst-case bound

$$|\langle \mathbf{h}(c), \mathbf{d}_f \rangle - \alpha_f(c)| \leq \mu(k-1)A + \varepsilon$$

follows. Projection onto the architectural readout direction $\mathbf{w}_f$ admits the same decomposition with an additional $\sqrt{2\gamma}\|\mathbf{h}(c)\|$ term from readout–dictionary misalignment.

The first bound is the main content: projection succeeds exactly when cross-feature interference $I_f(c)$ and residual contamination $R_f(c)$ are small. Sparsity and low-coherence are *one sufficient way* to make these terms small, yielding the worst-case $\mu(k-1)A + \varepsilon$ expression standard in compressed-sensing analyses; they are not the only way. Crucially, $I_f(c)$ and $R_f(c)$ are individually empirically measurable: the SAE overlap analysis in Section 4 estimates the cross-feature term by measuring the support overlap between class-prompt-active and instance-active SAE latents, and the wrong-class projection mass tracks how much $\langle \mathbf{h}(c), \mathbf{d}_j \rangle$ deviates from zero for $j \neq y$ on instances of class y (a direct estimate of $|I_j(c)|$ for non-active directions). The theorem therefore provides a diagnostic decomposition rather than a black-box assumption: when projection-based recovery succeeds or fails, the framework predicts which interference term is responsible.

Combining the architectural and statistical content gives a concrete account of next-token prediction. Expanding $\mathbf{u}_t^\top \mathbf{h}(c)$ over both decompositions, the matching part isolates the bundle features active in context:

$$\ell_t(c) = \mathbf{u}_t^\top \mathbf{h}(c) = \sum_{f \in B(t) \cap S(c)} b_{t,f} \alpha_f(c) + \Delta_t(c),$$

with $\Delta_t(c)$ bounded by the same interference terms as Theorem 3.3 (Proposition A.1, Appendix A). Next-token prediction reduces to weighted feature-bundle matching against the active features in the residual stream.

Corollary 3.4 (Additive Feature Contributions and Intervention Predictions). *Under the bundle-matching decomposition above, the contribution of feature f to logit $\ell_t(c)$ is $b_{t,f} \alpha_f(c)$; the contribution of a group $G \subseteq S(c)$ is $\sum_{f \in G \cap B(t)} b_{t,f} \alpha_f(c)$; and an intervention setting $\alpha_f(c) \rightarrow \alpha_f^*$ produces a logit change of*

$$\Delta \ell_t = b_{t,f} (\alpha_f^* - \alpha_f(c)) + \Delta_t^{\text{int}}(c),$$

where $\Delta_t^{\text{int}}(c)$ is bounded by the same interference terms of Theorem 3.3.

The corollary makes the framework’s predictions for causal-intervention experiments quantitative. Suppressing a feature f along $\mathbf{d}_f$ should change the logit margin by an amount predictable from $b_{t,f} \alpha_f(c)$; suppressing a group of features should change it by the sum of individual effects; and across instances of class y , larger $\alpha_y(c)$ should produce proportionally larger logit margins for y , with the slope set by the bundle coefficient $b_{y,y}$. The magnitude $\alpha_f(c)$ therefore operationalizes per-context feature importance directly: invariant directions $\mathbf{d}_f$ make the only context-dependent quantity the scalar magnitude, and downstream contribution is governed by the bundle coefficient of the consuming interface. We do not run intervention experiments in this paper, but the framework specifies what they should produce; the empirical evidence in Section 5 on projection-based recovery measures the same quantities that an intervention experiment would target.

3.3 The Self-Reference Property

A direct consequence is that class prompts can serve as approximate references for feature directions without supervision. A class prompt t_f —a token or expression canonically expressing feature f —is itself a context where f is active. Its activation, which we write $\mathbf{h}_{t_f}$, therefore decomposes as $\mathbf{h}_{t_f} = \beta_f \mathbf{d}_f + \boldsymbol{\nu}_f$ with $\beta_f > 0$ (the class-prompt magnitude) and $\|\boldsymbol{\nu}_f\| \leq \delta_f$ (residual contamination from other components active in the prompt). Up to the contamination ratio δ_f/β_f , the class-prompt activation is aligned with $\mathbf{d}_f$ and can be used in its place to detect f in any other context.

Theorem 3.5 (Self-Reference Classification Margin). *Let $\{\mathbf{d}_1, \dots, \mathbf{d}_K\}$ be unit class-feature directions with cross-class coherence $|\langle \mathbf{d}_i, \mathbf{d}_j \rangle| \leq \mu_C$ for $i \neq j$. Suppose the class prompt activation for class k decomposes as $\mathbf{h}_{t_k} = \beta_k \mathbf{d}_k + \boldsymbol{\nu}_k$ with $\beta_k > 0$ and $\|\boldsymbol{\nu}_k\| \leq \delta_k$, and an instance c of class y decomposes as $\mathbf{h}(c) = \alpha_y(c) \mathbf{d}_y + \boldsymbol{\eta}(c)$ with $\alpha_y(c) > \tau_y$ and $|\langle \mathbf{d}_k, \boldsymbol{\eta}(c) \rangle| \leq \tau_k$ for every k . Then the self-reference classifier $\hat{y}(c) = \arg \max_{k \in [K]} \mathbf{h}_{t_k}^\top \mathbf{h}(c)$ is correct if, for every $j \neq y$,*

$$\beta_y (\alpha_y(c) - \tau_y) - \delta_y \|\mathbf{h}(c)\| > \beta_j (\mu_C \alpha_y(c) + \tau_j) + \delta_j \|\mathbf{h}(c)\|.$$

Each term in the margin condition maps to a measurable quantity: $\alpha_y(c)$ is the class signal strength, μ_C is cross-class direction coherence, τ_j is instance interference along class j ’s direction, and δ_j is class-prompt contamination. Failure modes correspond to violations of specific terms (low signal, overlapping classes, contaminated prompts), each of which is empirically diagnosable. This is the

central theorem behind the experimental section: the class-prompt classifier is exactly the empirical instantiation of $\mathbf{h}_{t_k}^\top \mathbf{h}(c)$, and the SAE overlap analysis in Section 4 provides direct evidence on the μ_C and τ_j terms.

The mechanism behind Self-Reference. The Self-Reference Property has a mechanistic basis in how attention and the unembedding interact through the shared feature dictionary. When attention’s OV map approximately preserves dictionary directions, attention aggregates per-feature evidence from prompt tokens into the prediction position, where the unembedding scores candidates by feature-bundle matching. Cross-entropy then reinforces this coordination during training: the gradient with respect to a residual-stream feature magnitude is negative whenever the target token contains that feature more strongly than the model’s current predicted distribution does, so gradient descent increases the corresponding feature coordinate. We formalize the aggregation step (Proposition A.31) and the reinforcement step (Corollary A.32) in Appendix A. Together they describe a closed loop in which surface forms that express the same underlying feature—“France,” “Paris,” “Marseille,” “Eiffel Tower”—are routed through the same output-facing coordinate because they all support next tokens whose unembedding bundles include the corresponding feature component. Self-Reference is the observable endpoint: a class prompt elicits the feature directly, while semantically related contexts elicit it indirectly, and both work because the unembedding scores future tokens through the same coordinate.

3.4 Extension to MLP blocks

The framework extends to features written by MLP blocks under a structural condition we call a *selection-only feature decomposition*: the block’s output decomposes as $\sum_i g_i(\mathbf{h}) \mathbf{w}_i^{\text{out}}$, where each $\mathbf{w}_i^{\text{out}}$ is a fixed direction and each $g_i(\mathbf{h})$ depends only on its own read direction. A 2-layer MLP structurally satisfies this condition: columns of W_{out} are the fixed write directions, and the gate $g_i(\mathbf{h}) = \sigma(\langle \mathbf{w}_i^{\text{in}}, \mathbf{h} \rangle + b_{\text{in},i})$ depends only on the unit’s own read direction. Deeper MLPs ($L_{\text{out}} > 2$) retain fixed final write directions but break the independent-magnitude condition through compositional gating (Proposition A.27 in Appendix A). The picture therefore covers attention OV circuits, the unembedding, and 2-layer MLP blocks alike: each writes through fixed directions modulated by context-dependent scalar gates, and each can be analyzed by projection onto the corresponding feature directions.

Scope: features are domain-local, not global coordinates. The picture above is not a claim that each feature occupies a fixed global coordinate sitting everywhere in the residual stream. A feature f has a domain of expression—the set of contexts where it is active or needed—and the sparse-superposition decomposition $\mathbf{h}(c) = \sum_{g \in S(c)} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c)$ already encodes this: in a context c with $f \notin S(c)$, the feature is simply absent. Within a feature’s domain of expression, our claim is that the feature is communicated through its corresponding fixed direction $\mathbf{d}_f$ (or low-dimensional subspace), with context-dependent magnitude $\alpha_f(c)$; this is what allows circuit tracing and linear probes to recover the feature by projecting onto $\mathbf{d}_f$. Outside that domain, what happens to the residual capacity along $\mathbf{d}_f$ —whether it is unused, occupied by other features active in different contexts, or repurposed by other computations—is outside the scope of this paper. The unembedding W_U remains the limiting case among the model’s interfaces: it is the only *globally shared* output interface, applied identically to every prediction at every position, so any feature that systematically affects a logit margin must become accessible through the corresponding fixed contrast direction $\mathbf{u}_y - \mathbf{u}_j$ on the contexts where it is active. Linear interpretability methods succeed when they probe a feature on its domain of expression, where the feature is present and exposed through its fixed direction; the framework does not characterize feature behavior outside the domain.

4 Methods for Extracting Directions

The Self-Reference Property (Corollary A.29) implies that a classifier built on class-prompt directions should generalize to instance prompts that express the same feature, since both routes land on the same invariant direction. We compare three approaches that leverage this insight, all of which classify by projecting attention-head activations onto normalized class directions; they differ only in how directions are extracted. We evaluate each attention head independently.

Dataset	Classes	Instances/Class
Animals	6	50
Countries	5	39
Emotional Sentences	6	60
Literary Quotes	6	50
Cartoon Phrases	6	50
Languages	6	50
Fruits	4	50
Companies	4	50

Table 1: Summary of task-specific prompt datasets used for probing invariant feature directions across semantic, stylistic, linguistic, and affective domains.

Zero-Shot Probe. Use class-token hidden states as class directions, mean-centered to remove components shared across all class prompts (e.g., “being a country”):

$$\hat{\mathbf{d}}_k = \frac{\mathbf{h}_{t_k} - \bar{\mathbf{h}}}{\|\mathbf{h}_{t_k} - \bar{\mathbf{h}}\|}, \quad \bar{\mathbf{h}} = \frac{1}{K} \sum_{j=1}^K \mathbf{h}_{t_j}. \quad (1)$$

No training data required.

Unsupervised Probe. A lightweight linear transformation W trained on *class tokens only* (no instance labels) via a contrastive objective:

$$\mathcal{L} = - \sum_k \log \frac{\exp((W\mathbf{h}_{t_k})^\top \mathbf{c}_k / \tau)}{\sum_j \exp((W\mathbf{h}_{t_k})^\top \mathbf{c}_j / \tau)}. \quad (2)$$

The Self-Reference Property predicts W should generalize: if tokens and instances share invariant directions, orthogonalizing tokens automatically orthogonalizes instances.

Sparse Autoencoder Probe. We train sparse autoencoders (SAEs) on attention-head outputs [Cunningham et al., 2023, Bricken et al., 2023, Kissane et al., 2024], producing sparse latents $\mathbf{z} = \text{TopK}(W_{\text{enc}}\mathbf{h} + \mathbf{b}_{\text{enc}})$. SAEs are trained only on instance activations (no class tokens); we then check whether class-token latents and instance latents activate the same SAE dimensions.

Unified Classification. Letting $\phi(\cdot)$ denote the chosen transformation (mean-centering, W , or SAE encoder), classification uses the Identity-Projection operator with normalized class vectors:

$$\hat{k} = \arg \max_{k \in [K]} \phi(\mathbf{h}(c))^\top \hat{\phi}_k, \quad \hat{\phi}_k = \frac{\phi(\mathbf{h}_{t_k})}{\|\phi(\mathbf{h}_{t_k})\|}. \quad (3)$$

The query is not normalized, so projection magnitude reflects feature strength.

5 Experiments

5.1 Architectural validation: linear vs. nonlinear readouts

Proposition 3.2 predicts that fixed-direction structure is architecturally necessary under linear readouts and merely permissible under nonlinear ones. We test this directly. Prior work on grokking [Power et al., 2022] shows that transformers learning modular arithmetic develop Fourier representations—embeddings arranged in a circle where addition becomes rotation. Jiang et al. [2024] attribute this linear structure to training dynamics (cross-entropy plus gradient descent) regardless of architecture; our hypothesis is that the readout matters: linear readouts make Fourier structure recoverable across seeds, while nonlinear readouts permit alternative solutions.

We train a 2-layer transformer on modular division ($p = 97$) with an MLP classification head instead of the standard linear unembedding, and train a separate linear probe on the same hidden states (with gradients detached). The MLP head achieves $\sim 95\%$ validation accuracy, but the linear probe fails ($\sim 20\%$) on runs where the model has found a non-Fourier solution; Fourier analysis confirms the

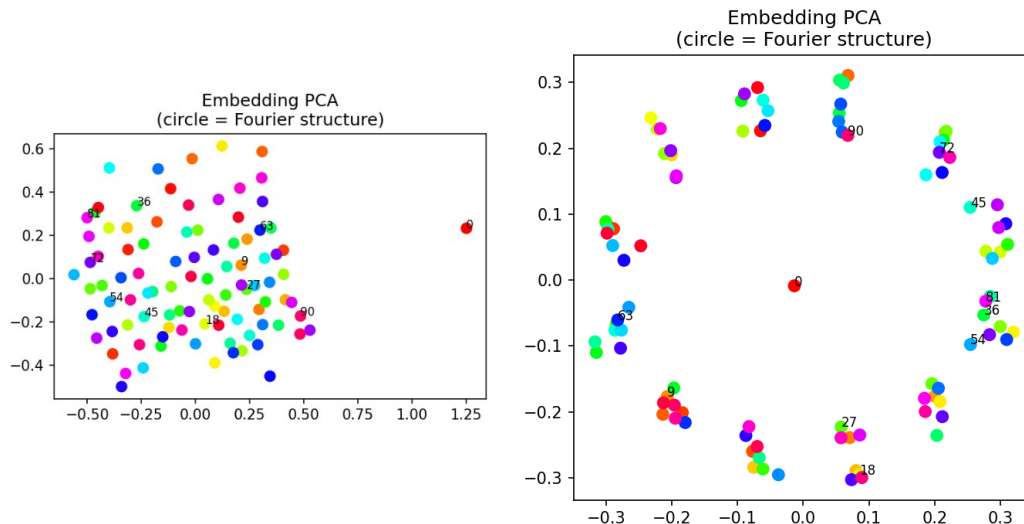


Figure 2: Linear readout layers constrain representation geometry. A 2-layer transformer is trained on modular division with an MLP classification head instead of linear unembedding. **(Left)** When the model finds a non-Fourier solution, embeddings lack circular structure and linear probes fail ($\sim 20\%$ accuracy). **(Right)** When the model discovers Fourier structure, linear probes succeed. Across random seeds, linear probe accuracy correlates with Fourier representation emergence: linear readouts make such structure necessary, while MLP heads make it merely possible.

embeddings lack circular structure (Fig. 2). Across random seeds, some runs do develop Fourier structure—and in exactly those runs, the linear probe also succeeds. This correlation demonstrates that Fourier representations are not inevitable under cross-entropy training: they emerge when the model happens to find that solution, but nonlinear readouts permit alternatives. Linear interfaces constrain representations to be directional; nonlinear interfaces permit but do not require such structure, exactly as Proposition 3.2 predicts.

We evaluate mainly across all the attention head outputs, on eight classification tasks spanning diverse semantic domains (Table 1): taxonomic (Animals), geographic (Countries), affective (Emotional Sentences [Ghazi et al., 2015]), stylistic (Literary Quotes, Cartoon Phrases), and linguistic (Languages [Artetxe et al., 2020]) and descriptive (Fruits, Companies).

For each task, we obtained a list of *implicit* sentences that express the class without mentioning it (e.g., “the country of the Eiffel Tower”). The full description of the datasets can be found in the Appendix F.1.

“The fruit associated with Newton’s discovery of gravity” $\rightarrow$ **Apple**

The Fruits and Companies datasets test behavior under polysemy (Appendix E.1). Our geometric analysis uses LLaMA3-8B, while classification results span four model families (Table 2) to demonstrate generalization.

5.2 Classification Results

Table 2 reports per-task classification accuracy. We highlight four findings.

Top-1 heads measure realization quality, not breadth. We report top-1 head accuracy because the framework predicts feature-specific readout directions in the heads that implement each feature, not uniform recoverability across all heads—many heads perform positional, syntactic, or copying computations that are not expected to encode any given semantic feature. Figure 4 answers the complementary breadth question: 86–98% of heads in LLaMA3-8B exhibit within-class alignment exceeding between-class alignment across four datasets.

Table 2: Classification accuracy (%) across methods and models. Top row reports SBERT cosine (all-mpnet-base-v2) as a label-free external-embedding baseline. Per-model rows report top-5 head mean accuracy for each method. Unsupervised values report mean and 95% CI over 5 seeds; SAE, Zero-Shot, and Text Output values are single-run. Supervised linear probes are reported in Appendix E.2 as an upper-bound calibration; the methods compared here all operate without labeled instance data. Text Output denotes a zero-shot conditional log-likelihood baseline scoring each class by token log-likelihoods given the prompt. The bottom block (*LLaMA3-8B (MLP)*) reports the same probes applied to MLP activations rather than attention-head outputs, at the last-token position only.

		Top-5 head mean Accuracy (%)							
Model	Method	Animals	Countries	C. Chars	Authors	Langs	Emotions	Fruits	Companies
SBERT		70.0	78.46	27.67	24.37	85.33	57.43	33.5	58.0
LLaMA3-8B	SAE	92.05	75.60	54.28	70.29	93.92	38.32	64.14	77.32
	Unsupervised	88.81 ± 0.19	87.41 ± 0.34	57.35 ± 0.25	70.15 ± 0.63	88.33 ± 0.40	63.77 ± 0.51	66.18 ± 0.26	78.74 ± 0.35
	Zero-Shot	84.22	82.97	54.46	62.95	89.12	52.43	59.36	80.26
	Text Output	99.67	79.48	42.00	68.75	94.00	68.40	49.00	88.50
Llama3.2-3B	SAE	78.81	82.43	53.09	49.29	92.63	35.76	61.26	72.63
	Unsupervised	86.09 ± 0.62	87.75 ± 0.35	53.81 ± 0.39	46.12 ± 0.43	79.45 ± 0.22	54.74 ± 0.08	52.40 ± 0.54	69.70 ± 0.51
	Zero-Shot	66.20	84.41	51.33	42.37	74.43	51.00	53.00	68.80
	Text Output	97.67	46.15	33.33	39.37	62.50	62.20	51.00	88.50
Mistral	SAE	82.10	74.48	49.95	52.69	97.62	43.64	56.52	75.72
	Unsupervised	87.0 ± 0.6	84.0 ± 1.2	54.0 ± 0.5	53.0 ± 0.7	98.0 ± 0.2	51.0 ± 0.3	64.0 ± 0.6	83.0 ± 0.3
	Zero-Shot	75.71	81.37	47.16	46.23	98.15	50.15	65.18	79.03
	Text Output	92.33	33.85	33.33	40.00	98.67	75.20	84.50	89.00
GPT2-Small	SAE	26.65	39.62	20.21	15.73	84.02	17.69	30.83	52.59
	Unsupervised	26.20 ± 0.90	38.34 ± 0.84	26.45 ± 0.58	19.62 ± 0.40	69.53 ± 1.52	28.32 ± 0.27	35.10 ± 0.77	51.02 ± 0.74
	Zero-Shot	25.39	37.62	20.83	20.11	88.11	17.37	33.90	29.06
	Text Output	35.33	25.64	17.00	12.50	15.67	49.80	37.50	29.50
LLaMA3-8B (MLP)	Unsupervised	77.20	73.40	59.60	73.90	48.20	21.00	35.40	71.40
	Zero-Shot	63.60	50.00	42.20	60.20	59.90	12.70	46.90	66.50

All three methods recover the predicted structure. Zero-shot, unsupervised, and SAE-based classification all distinguish between classes of similar semantic meaning, providing converging evidence across methods that differ only in how directions are extracted—directly from tokens, via learned orthogonalization, or through sparse decomposition. The unsupervised method consistently outperforms zero-shot, indicating that learned orthogonalization improves separability by disentangling shared category components (“being a country”) from class-specific ones.

SAEs recover the same invariant directions. SAE-based classification reaches competitive accuracy despite SAEs being trained only on instance activations, never on class tokens—the Identity-Projection operator therefore connects sparse-autoencoder interpretability to the framework, and the SAE-overlap analysis (Section 5.3, Fig. 3) provides direct evidence on the dictionary-realization condition (Remark A.28).

Generalization across architectures and modalities. Consistent performance across LLaMA3-8B, Mistral-7B, GPT2-Small, and LLaMA3.2-3B suggests directional invariance is not an artifact of any particular family or scale. The bottom block of Table 2 probes MLP activations at the last-token position: attention consistently outperforms MLP on categorical tasks (Animals, Countries, Languages, Emotions), but the gap narrows on more compositional or stylistic tasks (Authors, Characters, Companies). MLP outputs being decodable above chance is consistent with the framework: MLP outputs are added back to the residual stream and must be readable by downstream linear interfaces, so any MLP-produced feature that influences later computation must retain a linearly accessible component.

5.3 SAE-overlap analysis: tokens activate the same features as instances

To test whether class tokens and their semantic instances activate the *same* SAE features, we trained one sparse autoencoder per attention head following Kissane et al. [2024], using only instance activations—class tokens were never seen during training. Despite this, class-token activations and instance activations consistently activate overlapping top- k SAE dimensions. Figure 3 illustrates the animal dataset: with $k=32$, mammals share 15/32 features, fish 23/32, reptiles 25/32, birds 22/32. The pattern holds across datasets and model sizes, and provides direct evidence on the cross-feature interference term $I_f(c)$ in Theorem 3.3: the same dictionary atoms that organize residual activations across instances also organize the activations elicited by class prompts.

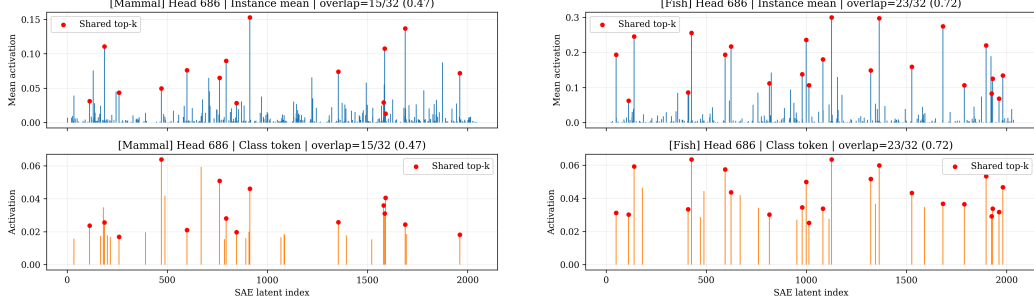


Figure 3: SAE shared-peak analysis. Top- k SAE dimensions of a class token compared with the top- k most frequent SAE dimensions across instances of that class. Red markers denote shared dimensions.

5.4 Activation Patching

To validate Corollary 3.4, we use the Zero-Shot Probe approach extended to two contrastive prompts and two classes. Given a clean and corrupted prompt pair $(p_{\text{clean}}, p_{\text{corr}})$ with corresponding answers $(a_{\text{clean}}, a_{\text{corr}})$, we extract the class-direction

$$\Delta a = h(a_{\text{clean}}) - h(a_{\text{corr}}) \in \mathbb{R}^{d_{\text{model}}} \quad (4)$$

from the residual stream at the last layer’s output. Since Δa lives in $\mathbb{R}^{d_{\text{model}}}$ but each component’s activation lives in its own native space, we project Δa into each component’s native space through the architecture’s output matrix— $W_O^{\ell, h}$ for attention head h at layer ℓ , W_{out}^{ℓ} for the MLP at layer ℓ :

$$d_{\ell, h}^{\text{comp}} = W_O^c \frac{\Delta a}{\|\Delta a\|} \in \mathbb{R}^d, \quad (5)$$

$$(6)$$

We then rank components by the alignment of their prompt-side activation difference with this back-projected direction:

$$s_{\ell, h}^{\text{attn}} = \langle z_{\ell, h}(p_{\text{clean}}) - z_{\ell, h}(p_{\text{corr}}), d_{\ell, h}^{\text{attn}} \rangle, \quad (7)$$

$$s_{\ell}^{\text{mlp}} = \langle m_{\ell}(p_{\text{clean}}) - m_{\ell}(p_{\text{corr}}), d_{\ell}^{\text{mlp}} \rangle, \quad (8)$$

where $z_{\ell, h}$ is the per-head output (in $\mathbb{R}^{d_{\text{head}}}$) and m_{ℓ} is the post-activation MLP vector (in $\mathbb{R}^{d_{\text{mlp}}}$). Finally, we patch the top- k components from clean to corrupted and measure how much of the logit-difference metric is recovered, comparing against gradient-based attribution patching (AtP) and its GradDrop variant (AtP*) as baselines.

5.5 Cross-Architecture Validation

The framework predicts invariant subspaces wherever a transformer communicates through linear interfaces, regardless of the training objective or input modality. To test this prediction, we evaluate three architectures outside the decoder-only LLM regime: BERT (encoder-only, masked language modeling), ViT (vision-only encoder), and LLaVA (vision-language). For BERT we apply the zero-shot probe to a subset of the text tasks. For ViT, which has no text vocabulary, we use the [CLS] representation of a canonical image as the class direction. For LLaVA we use object names as class directions and images as instances. ViT and LLaVA are evaluated on a 10-class, 1000-image ImageNet subset.

All five settings recover the invariant structure well above the random baseline (Table 4). BERT—an encoder-only model trained with masked language modeling rather than next-token prediction—reaches 93% on Languages, comparable to the decoder-only models in Table 2. The vision and vision-language results are more striking: ViT has no text interface at all, yet the [CLS] image representation provides a class direction that linearly separates ImageNet classes at roughly $8\times$ random. The refCOCO entry in Table 5 adds a further cross-modal data point, with LLaVA aligning textual descriptions to images at $5\times$ random. Together these results are consistent with the framework’s prediction that the constraint is architectural—linear interfaces require feature exposure in fixed directions—rather than language- or autoregression-specific.

Table 3: Attribution method comparison on GPT-2 small. Faithfulness (F) and completeness (C) at small k , where attribution methods matter most for circuit discovery, and AUC across $k \in \{1, 2, 3, 8, 16, 31, 78, 156\}$. On IOI (attention-heavy), SRP-BP recovers the circuit with 1–3 components, while AtP and SRP go negative due to attention saturation [?]. On Capitals (MLP-heavy factual recall), the layer-0 MLP dominates and AtP*, SRP-BP both identify it; the methods become indistinguishable.

Method	Faithfulness			Completeness		
	$k=1$	$k=3$	AUC	$k=1$	$k=3$	AUC
<i>IOI</i>						
AtP	−0.39	−0.15	0.971	−0.41	−0.16	0.965
AtP*	0.15	0.87	0.989	0.15	0.85	0.987
SRP	0.15	−0.15	0.961	0.15	−0.16	0.959
SRP-BP	0.87	1.12	1.000	0.82	1.07	0.997
<i>Capitals</i>						
AtP	0.30	0.96	0.982	0.35	0.97	0.983
AtP*	0.80	0.96	0.982	0.81	0.97	0.983
SRP	0.33	0.96	0.983	0.39	0.97	0.983
SRP-BP	0.80	0.89	0.985	0.81	0.89	0.983

Table 4: Cross-architecture validation. The Self-Reference Property holds across encoder-only (BERT), vision-only (ViT), and vision-language (LLaVA) transformers, supporting the prediction that linear interfaces—rather than autoregressive language modeling—are what drives invariant subspace structure.

Model	Modality	Task	Acc.	Random
BERT	text (encoder)	Languages	93.0%	16.7%
BERT	text (encoder)	Countries	61.5%	20.0%
BERT	text (encoder)	Emotions	48.9%	16.7%
ViT	vision	ImageNet-10	79.5%	10.0%
LLaVA	vision-language	ImageNet-10	78.6%	10.0%

5.6 Compositional and Multi-hop Tasks

To test whether the Self-Reference Property extends beyond categorical classification, we evaluate it on tasks requiring multi-hop reasoning, in-context binding, arithmetic composition, and cross-modal grounding (full task specifications in Appendix E.5). Probes substantially exceed random across all task families (Table 5), including refCOCO where the prototypes are textual descriptions and the instances are images. Two patterns are worth highlighting. First, the probe sometimes recovers the underlying operator even when the model fails to generate the correct output—most strikingly on language translation, where the probe reaches 58% while greedy generation only reaches 31%—indicating the directional structure exists internally even without successful downstream use. Second, performance degrades gracefully with hop depth ($k=1$ to $k=3$), tracking the model’s own generation accuracy, suggesting the probe reads what the model has actually computed rather than spurious surface cues.

6 Conclusion

We argued that transformer representations are organized as fixed feature directions whose magnitudes vary with context, and that linear interpretability methods recover meaningful structure because they read the same coordinates the architecture itself uses to make decisions. The architectural and statistical content is given by Propositions 3.1–3.2 and Theorem 3.3, with the Self-Reference Property (Corollary A.29) as a direct practical consequence: class prompts provide approximate references for feature directions in a single forward pass, with no labeled instance data or learned probes. Empirical validation across eight classification tasks, four model families, and three architectures confirms the predicted directional alignment between class prompts and instances. Linear interpretability is therefore neither a coincidence of training dynamics nor an artifact of probe choice, but a predictable consequence of how transformers *communicate* information internally—through fixed linear interfaces that constrain reused features to be exposed in linearly readable form.

Table 5: Compositional and multi-hop tasks. “Best Head” is the macro F1 of the best attention head; “Ratio” is best-head accuracy divided by the random baseline. Individual operation F1 reaches 0.89–1.0 in the ICL operator tasks.

Task	Cls.	Gen.	B. Head	Rand.	Ratio
Multi-hop chain ($k=1$)	8	79%	74%	12.5%	$5.9\times$
Multi-hop chain ($k=2$)	8	45%	39%	12.5%	$3.1\times$
Multi-hop chain ($k=3$)	8	35%	35%	12.5%	$2.8\times$
Language translation	5	31%	58%	20%	$2.9\times$
Vowel append	5	71%	61%	20%	$3.1\times$
Text operations	5	82%	48%	20%	$2.4\times$
Arithmetic (1-step add)	27	88%	86%	3.7%	$23\times$
Arithmetic (3-step add)	27	58%	66%	3.7%	$18\times$
Arithmetic (1-step mul)	27	92%	64%	3.7%	$17\times$
Arithmetic (2-step mul)	27	78%	56%	3.7%	$15\times$
Arithmetic (mul & add)	27	6%	24%	3.7%	$6.5\times$
SVAMP word problems	60	53%	45%	1.7%	$26\times$
refCOCO	10	—	50%	10%	$5\times$

Limitations. The framework characterizes features communicable through linear interfaces; features used only through nonlinear paths—routed through softmax QK with input-dependent attention weights, or composed through deeper MLPs that couple feature magnitudes (Proposition A.27)—are outside its scope. We do not characterize when invariant subspaces collapse to single directions versus higher-dimensional structures, and the in-distribution recoverability bounds do not extend to out-of-distribution compositional shifts where active feature sets or coherence structure change between training and deployment [Barin Pacela et al., 2026].

Future Directions. The Self-Reference Property suggests paths toward scalable, unsupervised circuit discovery, and the connection between SAE features and token directions offers new evaluation criteria for dictionary learning. The selection-only decomposition framing also opens a route to studying why circuit-tracing methods such as edge attribution patching succeed model-wide on standard transformers.

Impact Statement

This paper studies why linear interpretability methods recover stable structure from transformer representations. The work has dual-use implications. Constructively, a clearer account of when class-prompt directions are reliable supports model auditing, behavioral monitoring, safety-relevant feature discovery, and the evaluation of sparse autoencoders—all areas where unsupervised, label-free access to internal features is valuable for oversight. The same techniques can also lower the cost of identifying and manipulating specific behaviors via activation steering, including bypassing safety training. We do not propose any new attack technique, and the methods we describe are already implementable from public descriptions of activation steering and probing. We note explicitly that better tools for understanding model internals are also better tools for controlling them, and that responsible deployment of interpretability research includes considering who has access to the resulting techniques.

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576 A Detailed Theoretical Framework

577 This appendix provides the full theoretical framework underlying the compact main-body Section 3.
 578 We restate definitions and assumptions in detail, present supplementary corollaries and remarks
 579 (including the Identity-Projection operator and capacity-driven feature sharing), and discuss the
 580 bridges between the architectural readout direction $\mathbf{w}_f$ and the dictionary direction $\mathbf{d}_f$. All proofs of
 581 main-body results are given in Appendix B below.

582 A.1 Statement Restatements

583 For ease of reference, we restate here the main-body theorems and propositions whose detailed
 584 discussion appears below or whose proofs appear in Appendix B. The labels are duplicated on these
 585 restated versions so that appendix discussion and proofs can reference them naturally.

586 *Lemma* (Linear-Path Communication Lemma; restated). For a fixed linear interface $L : \mathbb{R}^d \rightarrow \mathbb{R}^{k_L}$
 587 and fixed downstream readout $\phi \in \mathbb{R}^{k_L}$, defining the residual-stream direction $\mathbf{w} := L^\top \phi$,

$$\phi^\top L\mathbf{h}(c) = \mathbf{w}^\top \mathbf{h}(c) \quad \forall c.$$

588 *Proposition* (Nonlinear Readouts Relax Output-Facing Directional Pressure; restated). Let logits be
 589 produced by a (possibly nonlinear) readout $\ell(c) = G(\mathbf{h}^{(L)}(c))$. Then

$$\nabla_{\mathbf{h}^{(L)}} \mathcal{L}(c) = J_G(\mathbf{h}^{(L)}(c))^\top (\mathbf{p}(c) - \mathbf{e}_y),$$

590 where J_G is the Jacobian of the readout. The rows of J_G depend on c for general nonlinear G . For
 591 a 2-layer MLP $G(\mathbf{h}) = W_{\text{out}}\sigma(W_{\text{in}}\mathbf{h} + \mathbf{b})$, the gradient simplifies to a context-gated sum over the
 592 fixed columns of $W_{\text{in}}^\top$:

$$\nabla_{\mathbf{h}^{(L)}} \mathcal{L} = W_{\text{in}}^\top \text{diag}(\sigma'(\cdot)) W_{\text{out}}^\top (\mathbf{p}(c) - \mathbf{e}_y).$$

593 For deeper readouts, the Jacobian becomes a chain product with cross-coupled gates; see Remark A.9.

594 *Proposition* (Next-Token Prediction as Feature-Bundle Matching; restated). Under Assumptions A.12
 595 and A.17, the logit assigned to token t in context c is

$$\ell_t(c) = \mathbf{u}_t^\top \mathbf{h}(c) = \sum_{f \in B(t) \cap S(c)} b_{t,f} \alpha_f(c) + \Delta_t(c),$$

596 where the interference term satisfies

$$|\Delta_t(c)| \leq \mu \sum_{f \in B(t)} |b_{t,f}| \sum_{\substack{g \in S(c) \\ g \neq f}} |\alpha_g(c)| + \|\mathbf{r}_t\| \|\mathbf{h}(c)\| + \|\mathbf{u}_t\| \|\boldsymbol{\xi}(c)\|.$$

597 *Example 1* (Solvable without a linear feature direction). Let representations be $\mathbf{h} = (x_1, x_2) \in$
 598 $\{-1, 1\}^2$ and the target be $y = x_1 x_2$. The two classes are not linearly separable in $\mathbf{h}$, so no fixed
 599 direction $\mathbf{v}$ and scalar b recover y via projection $\mathbf{v}^\top \mathbf{h} + b$. A 2-layer nonlinear readout, however,
 600 can compute y exactly. Cross-entropy training of such a readout can therefore be minimized without
 601 producing a linear feature for y in the input representation $\mathbf{h}$. This is the simplest concrete instance
 602 of the non-necessity content of Proposition 3.2: a learnable mapping from inputs to a correct output
 603 that does not factor through any fixed contrast coordinate in the upstream representation.

604 A.2 Preliminaries and Assumptions

605 We consider decoder-only transformers with the following properties:

606 **Assumption A.1** (Architectural Requirements). Let $\mathcal{M}$ be a transformer with L layers and hidden
 607 dimension d . We assume:

- 608 1. **Additive Residual Stream.** The residual stream updates additively: $\mathbf{r}^{(\ell+1)} = \mathbf{r}^{(\ell)} + \Delta^{(\ell)}$,
 609 where $\Delta^{(\ell)}$ is the output of layer ℓ .
- 610 2. **Linear Communication Interfaces.** The OV circuit in attention ($W_O W_V$), the QKV
 611 projections (W_Q, W_K, W_V), and the unembedding layer (W_U) act as linear maps on the
 612 residual stream.

613 3. **Shared Parameters.** The same parameters are applied across all token positions; optimization occurs over parameters shared across contexts.

614 4. **Linear Output Layer.** Token logits are computed via linear projection: $\text{logits} = W_U \mathbf{r}^{(L)}$.

616 These assumptions hold for standard transformer architectures including GPT-2, LLaMA, and similar models.

617 We denote by $\mathcal{L}$ the collection of read-side linear interfaces in the model:

$$\mathcal{L} = \{W_U\} \cup \{W_Q^{(\ell,h)}, W_K^{(\ell,h)}, W_V^{(\ell,h)}\}_{\ell,h} \quad (9)$$

619 indexed over layers ℓ and attention heads h . Each $L \in \mathcal{L}$ is a linear map $L : \mathbb{R}^d \rightarrow \mathbb{R}^{k_L}$ that is fixed at inference. The set $\mathcal{L}$ formalizes what we mean by “communication interfaces”: every linear map that reads from the residual stream and is shared across positions and contexts. We treat these as fixed maps that determine the *directions* along which content is read; the input-dependent dynamics of attention routing (the softmax of $Q^\top K / \sqrt{d}$) determine *how much* content is routed but do not modify these directions, and lie outside the scope of the feature-direction analysis.

625 A.3 Formal Definitions

626 **Definition A.2** (Context). A *context* $c = (x_1, \dots, x_n; i)$ consists of an input token sequence and a position of interest i . We write $\mathbf{h}^{(\ell)}(c) \in \mathbb{R}^d$ for the hidden state at layer ℓ and position i in context c .

628 **Definition A.3** (Semantic Feature). A *semantic feature* is a function $f : \mathcal{C} \rightarrow \mathcal{Y}$ mapping contexts to a value space $\mathcal{Y}$. For classification tasks, $\mathcal{Y} = \{y_1, \dots, y_K\}$ is finite.

630 **Definition A.4** (Communicable Feature). A semantic feature f is *communicable through interface* $L \in \mathcal{L}$ along direction $\phi \in \mathbb{R}^{k_L}$ if it satisfies both:

632 1. **Multi-context:** There exist distinct contexts c_1, c_2 with $f(c_1) = f(c_2)$.

633 2. **Linear decodability:** The feature value is recoverable through L :

$$\phi^\top L \mathbf{h}(c) = g(f(c)) \quad \forall c \in \mathcal{C} \quad (10)$$

634 for some function $g : \mathcal{Y} \rightarrow \mathbb{R}$. Equivalently, $\mathbf{w}_f^\top \mathbf{h}(c) = g(f(c))$ where $\mathbf{w}_f := L^\top \phi \in \mathbb{R}^d$ is the *decoding direction* in the residual stream.

636 The multi-context requirement captures that meaningful features appear across different surface realizations (e.g., “France” and “the country of the Eiffel Tower” both express the France feature). Linear decodability through $L \in \mathcal{L}$ captures that the feature must be extractable by some downstream linear interface—most importantly the unembedding W_U and the attention QKV projections.

640 **Definition A.5** (Invariant Subspace). A communicable feature f exhibits *contextual invariance* if there exists a subspace $\mathcal{S}_f \subseteq \mathbb{R}^d$, determined by f and model parameters alone, such that for all contexts c expressing f , the f -relevant information in $\mathbf{h}(c)$ lies entirely within $\mathcal{S}_f$.

643 **Definition A.6** (Directional Invariance). A communicable feature f exhibits *directional invariance* if it exhibits contextual invariance with $\dim(\mathcal{S}_f) = 1$, i.e., $\mathcal{S}_f = \text{span}(\{\mathbf{d}_f\})$ for some direction $\mathbf{d}_f \in \mathbb{R}^d$.

646 A.4 Main Theoretical Results

647 The remainder of this section gives the architectural and statistical content of this picture. The architectural content is twofold. First, the Linear-Path Communication Lemma (Lemma A.1) records the elementary identity that fixed downstream linear paths correspond to fixed residual-stream directions. Second, Proposition 3.1 traces this identity into output decisions and training: every logit margin between two output tokens equals the projection of the final residual stream onto a fixed contrast direction, and the cross-entropy gradient pushes representations along weighted combinations of the same contrast directions. Proposition A.1 sharpens the architectural specificity: a 2-layer MLP readout preserves fixed read directions through context-dependent scalar gates, while deeper compositions generically break this independent-gating factorization. The statistical content adds the conditions under which the picture is recoverable from data: sparse low-coherence feature

organization (Assumption A.12) and bounded readout–dictionary alignment (Assumption A.13) yield Theorem 3.3, which bounds the error of projection-based feature recovery. Combining these with a token-bundle decomposition of the unembedding (Assumption A.17) gives Proposition A.1 and Corollary A.18: next-token prediction reduces to weighted feature-bundle matching, and contributions to logit margins are read through fixed contrast directions. The Self-Reference Property follows: class prompts, being themselves contexts that express a feature, provide approximate references for its direction without supervision (Corollary A.29), with explicit margin conditions for the resulting classifier (Theorem 3.5).

The lemma’s proof is one line, and that is the right shape for a load-bearing identity: the architectural fact it records (fixed linear paths induce fixed directions) is what the rest of the framework leverages. The first such leveraging is at training time, through the cross-entropy gradient.

Interpretation: architecture and training share the same coordinates. The proposition has a sharper content than “the gradient lies in some subspace.” It says that the same fixed contrast directions $\mathbf{u}_y - \mathbf{u}_j$ govern both how the model decides between two tokens at inference and how training updates push the residual stream during learning. Cross-entropy is not arbitrary pressure: it rewards making the evidence for the correct token accessible through the same contrast coordinates the model uses to compare tokens. This connects optimization directly to the geometry of decisions, without requiring a separate assumption about what features are linearly decodable.

Remark A.7 (Scope: final residual stream, not arbitrary layers). Proposition 3.1 is stated for the gradient at the *final* residual stream $\mathbf{h}^{(L)}$, where the cross-entropy loss is applied directly through W_U . Earlier-layer gradients are obtained by backpropagating through subsequent attention, MLP, and normalization layers, picking up additional Jacobians along the way:

$$\nabla_{\mathbf{h}^{(e)}} \mathcal{L} = J_{\ell \rightarrow L}(\mathbf{h}^{(\ell)})^\top W_U^\top (\mathbf{p} - \mathbf{e}_y).$$

The fixed-direction pressure is therefore a property of the output-facing path. Earlier layers may maintain intermediate features in different geometric arrangements; for those features to affect the next-token distribution, however, they must eventually be converted into output-facing components readable by the unembedding.

Remark A.8 (From training pressure to learned representations). Proposition 3.1 characterizes the gradient *pressure* on $\mathbf{h}^{(L)}$, not the learned representation directly. The standard implicit assumption of any post-hoc analysis of trained networks is that representations have, after sufficient training, accumulated and aligned with the directions of repeated gradient pressure. Under this assumption, output-relevant features are exposed through the contrast directions $\mathbf{u}_y - \mathbf{u}_j$ that govern token decisions, and Theorem 3.3 below characterizes when projection onto these directions recovers the underlying feature magnitudes.

Remark A.9 (Why deeper readouts break independent-direction pressure). For a readout with $L_{\text{out}} > 2$ layers, the Jacobian becomes a chain product

$$J_G(\mathbf{h}) = W_{L_{\text{out}}} \prod_{l < L_{\text{out}}} \text{diag}(\sigma'_l) W_l,$$

in which each $\text{diag}(\sigma'_l)$ depends on the activations at layer l , which themselves depend on earlier gates. The effective gradient direction at any context is no longer a per-context-gated sum over a fixed set of column directions; it is a context-dependent linear combination *of* such sums. Feature pressures become coupled across readout layers, and the selection-only structure preserved by 2-layer MLPs no longer holds. This matches the empirical observation in Section 5.1: under linear and 2-layer MLP readouts, linear (Fourier) structure appears reliably across seeds, while under deeper nonlinear readouts it appears only intermittently.

Why shared directions are an efficient solution (full version). Propositions 3.1 and 3.2 establish what is and is not architecturally required. They do not by themselves explain why training tends to produce shared, reusable feature directions across many contexts. The following proposition fills that gap under an efficiency bias: among all representations satisfying a margin requirement through a linear readout, the minimum-norm solution is the one in which all such contexts share a single direction with magnitudes scaled by the required margin.

Proposition A.10 (Norm-Efficient Solutions Collapse onto Fixed Contrast Directions). *Fix a contrast direction $\mathbf{v} := \mathbf{u}_y - \mathbf{u}_j$, and let $\{c_i\}_{i=1}^N$ be contexts each requiring margin $m_i > 0$ on the (y, j)*

707 *contrast. For each context, the minimum-norm representation that achieves the required margin*
 708 *under the linear readout,*

$$\mathbf{h}_i^* = \arg\min_{\mathbf{h} \in \mathbb{R}^d} \|\mathbf{h}\|^2 \quad \text{subject to} \quad \mathbf{v}^\top \mathbf{h} \geq m_i,$$

709 *is*

$$\mathbf{h}_i^* = \frac{m_i}{\|\mathbf{v}\|^2} \mathbf{v}.$$

710 *All $\mathbf{h}_i^*$ lie on the line spanned by $\mathbf{v}$, with magnitudes proportional to the required margin: same*
 711 *direction, context-dependent magnitude.*

712 *Proof.* Decompose $\mathbf{h} = a\mathbf{v} + \mathbf{r}$ with $\mathbf{r} \perp \mathbf{v}$. The constraint $\mathbf{v}^\top \mathbf{h} = a\|\mathbf{v}\|^2 \geq m_i$ requires
 713 $a \geq m_i/\|\mathbf{v}\|^2$, and the norm $\|\mathbf{h}\|^2 = a^2\|\mathbf{v}\|^2 + \|\mathbf{r}\|^2$ is minimized at $\mathbf{r} = 0$ and $a = m_i/\|\mathbf{v}\|^2$. $\square$

714 *Remark A.11 (Nonlinear readouts do not generally collapse).* For a nonlinear margin $m_{y,j}(\mathbf{h}) =$
 715 $G_y(\mathbf{h}) - G_j(\mathbf{h})$, the corresponding minimum-norm problem replaces the linear constraint with
 716 $m_{y,j}(\mathbf{h}_i) \geq m_i$. At an active KKT point, $\mathbf{h}_i^*$ is parallel to $\nabla_{\mathbf{h}} m_{y,j}(\mathbf{h}_i^*)$, which depends on $\mathbf{h}_i^*$ for
 717 nonlinear G . Different contexts can therefore have different optimal directions; collapse to one shared
 718 direction occurs only under additional structure (e.g., affine $m_{y,j}$, or parallel margin gradients on the
 719 relevant domain). The non-collapse here parallels the non-necessity of Proposition 3.2.

720 **Scope of the efficiency claim.** Proposition A.10 characterizes the optimum of an explicit norm-
 721 minimization problem with one contrast direction. Real transformers do not solve this problem exactly,
 722 but several standard mechanisms produce a similar efficiency bias: weight decay penalizes parameter
 723 norms, gradient descent on classification objectives has a well-documented implicit bias toward small-
 724 norm solutions, and LayerNorm bounds activation norms at every block. The proposition formalizes
 725 what such an efficiency bias *prefers* when applied through a linear readout: contexts requiring the
 726 same contrast collapse onto a shared direction. With many contrast directions simultaneously, this
 727 composes into the picture the paper develops throughout: each output-relevant feature occupies a
 728 fixed direction (or low-dimensional subspace) shared across the contexts that use it. The grokking
 729 ablation in Section 5.1 provides empirical support: under linear readouts, this directional structure
 730 emerges consistently across seeds; under deeper nonlinear readouts—where Remark A.11 says no
 731 such collapse is required—the structure emerges only intermittently. We do not claim this proposition
 732 derives feature emergence; emergence depends on optimization dynamics, training data, and inductive
 733 biases that existing work studies. The proposition characterizes the geometric *form* that an efficient
 734 solution takes when one is found.

735 **Assumption A.12 (Sparse Feature Dictionary).** Let $\mathcal{D} = \{\mathbf{d}_g\}_{g \in \mathcal{F}} \subset \mathbb{R}^d$ be a dictionary of unit-norm
 736 feature directions, $\|\mathbf{d}_g\| = 1$, with coherence

$$\mu := \max_{g \neq g'} |\langle \mathbf{d}_g, \mathbf{d}_{g'} \rangle|.$$

737 Assume that each context c admits a sparse feature decomposition

$$\mathbf{h}(c) = \sum_{g \in S(c)} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c),$$

738 where $S(c) \subseteq \mathcal{F}$ is the set of active features, $|S(c)| \leq k$, and $\|\boldsymbol{\xi}(c)\| \leq \varepsilon$. We say that context c
 739 *expresses* feature f when $f \in S(c)$ and $\alpha_f(c) \neq 0$.

740 This assumption formalizes the superposition hypothesis [Elhage et al., 2022]: features are en-
 741 coded along an overcomplete dictionary of directions, and only a small number are active in any
 742 given context. We use $\mathbf{d}_f$ for the feature direction in the sparse-superposition analysis below; the
 743 bounds characterize when projection onto this direction is reliable. Whether $\mathbf{d}_f$ coincides with the
 744 architectural readout direction $\mathbf{w}_f$ from Proposition 3.1 is a separate identifiability question—in
 745 non-orthogonal dictionaries, the optimal linear readout for a feature can be closer to a dual vector
 746 than to the dictionary atom. We address this gap explicitly with a bounded-misalignment assumption.

747 **Assumption A.13 (Readout–Dictionary Alignment).** For each architecturally communicable feature
 748 f that also appears in the dictionary $\mathcal{D}$, the architectural readout direction $\mathbf{w}_f$ from Proposition 3.1
 749 and the dictionary direction $\mathbf{d}_f$ from Assumption A.12 satisfy

$$\frac{\langle \mathbf{w}_f, \mathbf{d}_f \rangle}{\|\mathbf{w}_f\| \|\mathbf{d}_f\|} \geq 1 - \gamma$$

for some *readout misalignment* parameter $\gamma \in [0, 1)$. Equivalently, the angle between the two directions is at most $\arccos(1 - \gamma)$.

The scalar $\alpha_f(c)$ in the dictionary decomposition is, up to a multiplicative constant absorbed into the dictionary normalization, the same quantity as the readout output $g(f(c))$ from the proposition. The Theorem below is stated in terms of projections onto $\mathbf{d}_f$ for clarity; the corollary that follows shows projections onto $\mathbf{w}_f$ recover the same quantity up to an additional $O(\sqrt{\gamma})$ term.

Corollary A.14 (Readout–Dictionary Bridge). *Under Assumptions A.12 and A.13, projection onto the architectural readout direction $\mathbf{w}_f$ recovers the same feature magnitude as projection onto the dictionary direction $\mathbf{d}_f$, up to a misalignment-controlled additional term:*

$$\left| \frac{\langle \mathbf{h}(c), \mathbf{w}_f \rangle}{\|\mathbf{w}_f\|} - \langle \mathbf{h}(c), \mathbf{d}_f \rangle \right| \leq \sqrt{2\gamma} \|\mathbf{h}(c)\|.$$

Combined with Theorem 3.3, this gives

$$\left| \frac{\langle \mathbf{h}(c), \mathbf{w}_f \rangle}{\|\mathbf{w}_f\|} - \alpha_f(c) \right| \leq \mu(k-1)A + \varepsilon + \sqrt{2\gamma} \|\mathbf{h}(c)\|.$$

Proof. Let $\hat{\mathbf{w}}_f := \mathbf{w}_f / \|\mathbf{w}_f\|$ be the unit readout direction. By Assumption A.13, $\langle \hat{\mathbf{w}}_f, \mathbf{d}_f \rangle \geq 1 - \gamma$, so $\|\hat{\mathbf{w}}_f - \mathbf{d}_f\|^2 = 2 - 2\langle \hat{\mathbf{w}}_f, \mathbf{d}_f \rangle \leq 2\gamma$ (using $\|\hat{\mathbf{w}}_f\| = \|\mathbf{d}_f\| = 1$). Therefore by Cauchy–Schwarz,

$$|\langle \mathbf{h}(c), \hat{\mathbf{w}}_f \rangle - \langle \mathbf{h}(c), \mathbf{d}_f \rangle| = |\langle \mathbf{h}(c), \hat{\mathbf{w}}_f - \mathbf{d}_f \rangle| \leq \|\mathbf{h}(c)\| \|\hat{\mathbf{w}}_f - \mathbf{d}_f\| \leq \sqrt{2\gamma} \|\mathbf{h}(c)\|.$$

The combined bound follows by triangle inequality with Theorem 3.3. $\square$

Corollary A.15 (Projection–Based Detection). *Under the assumptions of Theorem 3.3, define the interference radius*

$$\rho(c) := \mu \sum_{g \in S(c), g \neq f} |\alpha_g(c)| + \varepsilon.$$

If c expresses f with $|\alpha_f(c)| > \rho(c)$, then $|\langle \mathbf{h}(c), \mathbf{d}_f \rangle| > 0$. Moreover, if non-expressing contexts satisfy $\alpha_f(c) = 0$, then a threshold $\tau > \rho(c)$ separates contexts with sufficiently large f -magnitude from contexts in which f is absent.

Proof. Theorem 3.3 gives $\langle \mathbf{h}(c), \mathbf{d}_f \rangle = \alpha_f(c) + e_f(c)$ with $|e_f(c)| \leq \rho(c)$. If $|\alpha_f(c)| > \rho(c)$, the projection cannot vanish. The threshold statement follows from the same bound. $\square$

Corollary A.16 (Margin Condition for Multiclass Classification). *Consider a K -class task represented as one-vs-rest class indicators with directions $\{\mathbf{d}_1, \dots, \mathbf{d}_K\}$. Let c have true class y , and define the projection classifier*

$$\hat{y}(c) = \arg \max_{j \in [K]} \langle \mathbf{h}(c), \mathbf{d}_j \rangle.$$

Suppose each class projection has interference at most ρ , i.e., $|\langle \mathbf{h}(c), \mathbf{d}_j \rangle - \alpha_j(c)| \leq \rho$ for all $j \in [K]$. If the true class-feature margin satisfies $\alpha_y(c) - \max_{j \neq y} \alpha_j(c) > 2\rho$, then $\hat{y}(c) = y$.

Proof. For the true class y : $\langle \mathbf{h}(c), \mathbf{d}_y \rangle \geq \alpha_y(c) - \rho$. For any incorrect class $j \neq y$: $\langle \mathbf{h}(c), \mathbf{d}_j \rangle \leq \alpha_j(c) + \rho \leq \max_{j' \neq y} \alpha_{j'}(c) + \rho$. The score for y exceeds that of every other class whenever $\alpha_y(c) - \rho > \max_{j \neq y} \alpha_j(c) + \rho$, which is the stated margin condition. $\square$

Assumption A.17 (Token Feature Bundles). For the output tokens considered in a task, assume the unembedding row $\mathbf{u}_t = (W_U)_{t,:}$ admits an approximate sparse decomposition over the same feature dictionary $\mathcal{D} = \{\mathbf{d}_f\}_{f \in \mathcal{F}}$ used to decompose residual-stream activations in Assumption A.12:

$$\mathbf{u}_t = \sum_{f \in B(t)} b_{t,f} \mathbf{d}_f + \mathbf{r}_t.$$

Here $B(t) \subseteq \mathcal{F}$ is the token’s feature bundle, $b_{t,f} \in \mathbb{R}$ is the contribution of feature f to token t , and $\mathbf{r}_t \in \mathbb{R}^d$ is the unmodeled residual with $\|\mathbf{r}_t\| \leq \rho_t$. The bundle support sizes $|B(t)|$ are assumed sparse on the same order as the residual-stream support sizes $|S(c)|$.

This assumption is consistent with empirical findings on superposition in transformer matrices [Elhage et al., 2022]: just as the residual stream is modeled as a sparse combination of dictionary atoms, the unembedding rows are expected to be sparse combinations of the same atoms. We do not require this decomposition to hold for every token in the vocabulary, only for those participating in the task at hand.

The bundle-matching proposition gives a concrete architectural interpretation of next-token prediction: the unembedding scores each candidate token by how well its feature bundle matches the active features currently encoded in the residual stream. This sharpens the architectural argument: every token whose probability the model can affect must have its bundle features exposed in directions the unembedding can read, because the unembedding has no other way to score them. The Self-Reference Property below follows directly: a class prompt’s unembedding row is itself a feature bundle for that class, so its activation aligns with the same dictionary directions as any other context expressing those features.

Corollary A.18 (Feature Contributions to Logit Margins). *Let y and j be two output tokens with unembedding rows $\mathbf{u}_y$ and $\mathbf{u}_j$, and define the contrast direction*

$$\mathbf{v}_{y,j} := \mathbf{u}_y - \mathbf{u}_j.$$

Suppose $\mathbf{v}_{y,j}$ decomposes over the feature dictionary as

$$\mathbf{v}_{y,j} = \sum_{f \in B(y,j)} \beta_{y,j,f} \mathbf{d}_f + \mathbf{r}_{y,j},$$

with $\|\mathbf{r}_{y,j}\| \leq \rho_{y,j}$. Then for a context activation $\mathbf{h}(c) = \sum_{g \in S(c)} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c)$ as in Assumption A.12, the logit margin satisfies

$$\ell_y(c) - \ell_j(c) = \sum_{f \in B(y,j) \cap S(c)} \beta_{y,j,f} \alpha_f(c) + \Delta_{y,j}(c),$$

where the interference term satisfies

$$|\Delta_{y,j}(c)| \leq \mu \sum_{f \in B(y,j)} |\beta_{y,j,f}| \sum_{\substack{g \in S(c) \\ g \neq f}} |\alpha_g(c)| + \|\mathbf{r}_{y,j}\| \|\mathbf{h}(c)\| + \|\mathbf{v}_{y,j}\| \|\boldsymbol{\xi}(c)\|.$$

A feature f supports token y over j when $\beta_{y,j,f} > 0$ and opposes it when $\beta_{y,j,f} < 0$. Thus, features contribute to the model’s token choice through fixed output-facing contrast directions, up to bounded sparse-superposition interference.

Proof. By linearity, $\ell_y(c) - \ell_j(c) = (\mathbf{u}_y - \mathbf{u}_j)^\top \mathbf{h}(c) = \mathbf{v}_{y,j}^\top \mathbf{h}(c)$. Applying Proposition A.1 with the contrast direction $\mathbf{v}_{y,j}$ in place of a single unembedding row $\mathbf{u}_t$ yields the matching term and the stated interference bound; the bundle support $B(y,j)$ collects all features that distinguish the two tokens, and $\mathbf{r}_{y,j}$ is the unmodeled component of the contrast. $\square$

This corollary connects the framework directly to model decisions: classification between alternatives is governed by feature-weighted contrast scores, with each feature’s contribution determined by how it appears in the relevant unembedding contrast direction. It also explains a class of empirically observed phenomena: features that distinguish closely related tokens (e.g., singular vs. plural, masculine vs. feminine, member-of-class vs. not-member) should appear with opposing signs in $\mathbf{v}_{y,j}$, and the magnitude of $\beta_{y,j,f}$ predicts the strength of that feature’s effect on token choice.

Remark A.19 (Multiclass features). For multiclass datasets, we do not assume that the entire K -valued class variable occupies a single one-dimensional subspace. Instead, we treat the task as a collection of one-vs-rest class indicators. The full class variable may occupy a higher-dimensional class subspace (Proposition C.1 in the appendix shows $\dim(\mathcal{S}_f) \leq K - 1$), while each class indicator is associated with its own directional readout. This is the setting used by the projection classifier in Corollary A.16 and by the class-token probes in Section 4.

Remark A.20 (Compatibility with the $\alpha_f \mathbf{d}_f + \boldsymbol{\eta}$ notation). Throughout the rest of the paper we sometimes write the decomposition compactly as $\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \boldsymbol{\eta}(c)$, where $\boldsymbol{\eta}(c) := \sum_{g \in S(c), g \neq f} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c)$ collects all non- f contributions. We do not require $\boldsymbol{\eta}(c) \perp \mathbf{d}_f$: under superposition, $\boldsymbol{\eta}(c)$ generally has nonzero overlap with $\mathbf{d}_f$, and Theorem 3.3 bounds exactly this overlap.

Note on basis-dependence. Residual stream geometry is non-canonical: applying any invertible $A \in \mathbb{R}^{d \times d}$ to the residual stream and absorbing A^{-1} into all weight matrices yields a behaviorally equivalent model with different-looking “directions” in coordinates. The invariant direction $\mathbf{d}_f$ is therefore not a basis-independent geometric object in absolute terms; Park et al. [2024] address this concern formally via causal inner products. The framework’s claim is more restricted but, we argue, sufficient for its purpose: *within a fixed parameterization, all activations and weight matrices share the same basis*. Class-prompt activations $\mathbf{h}_t$ and instance activations $\mathbf{h}(c)$ both inhabit the same residual-stream coordinates as the model that produced them, so projecting one onto the other is well-defined regardless of basis choice. If A transforms the residual stream, both $\mathbf{h}_t$ and $\mathbf{h}(c)$ transform together, and the projection $\mathbf{h}_t^\top \mathbf{h}(c)$ is invariant up to the metric implied by A . Identifiability concerns become substantive only when comparing directions *across* parameterizations (e.g., across models, training runs, or analyses that fix one space and vary another); they do not affect the framework’s intended use of within-model self-reference.

A.5 Capacity Constraints and Feature Sharing

The architectural argument so far identifies fixed decoding directions for features that are linearly communicable through some interface in $\mathcal{L}$. A complementary question is why features should be *factorized* into a small number of shared directions in the first place, rather than each token occupying its own dedicated direction. The vocabulary size $|\mathcal{V}|$ greatly exceeds the residual stream dimension d in practical transformers, which means most tokens must share structure with others. Under standard assumptions about feature sparsity and co-occurrence, this capacity constraint favors factorized representations: a small set of shared feature directions, with each token expressed as a sparse combination. We give a detailed argument in Appendix C; the conclusion connects directly to the superposition hypothesis [Elhage et al., 2022], with each shared factor satisfying the conditions of Theorem 3.3.

Remark A.21 (Implicit Classification). From this view, predicting a token *is* implicit classification: the model checks “does this context encode the factor combination for token t ?” Each factor $\mathbf{d}_f$ partitions contexts into those expressing f versus those that do not. Compression pressure from the capacity constraint ensures factors are reusable across tokens, maximizing the number of features that can coexist in the representation, which aligns with the information bottleneck principle [Tishby and Zaslavsky, 2015].

The decomposition $\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \boldsymbol{\eta}(c)$ from Theorem 3.3 is central to the framework: the direction $\mathbf{d}_f$ is invariant across contexts, while the *magnitude* $\alpha_f(c)$ varies. Linear interfaces preserve this structure:

$$W(\alpha \cdot \mathbf{d}_f) = \alpha \cdot (W\mathbf{d}_f) \quad (11)$$

A natural concern is whether LayerNorm—an input-dependent normalization applied before each block—disrupts directional invariance. It does not, in the following sense:

Proposition A.22 (LayerNorm preserves directional invariance). *Let $\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \boldsymbol{\eta}(c)$ where $\mathbf{d}_f \in \mathbb{R}^d$ is a context-invariant feature direction and $\alpha_f(c) \in \mathbb{R}$ is a context-dependent magnitude. If $\mathbf{d}_f \notin \text{span}(\mathbb{1})$, then there exists a context-invariant direction $\tilde{\mathbf{d}}_f$ such that*

$$\text{LN}(\mathbf{h}(c)) = \tilde{\alpha}_f(c) \tilde{\mathbf{d}}_f + \tilde{\boldsymbol{\eta}}(c) + \boldsymbol{\beta},$$

where $\tilde{\alpha}_f(c) := \alpha_f(c) / \sigma(\mathbf{h}(c))$ and $\boldsymbol{\beta}$ is the learned LayerNorm bias.

LayerNorm maps a context-invariant direction $\mathbf{d}_f$ to a transformed context-invariant direction $\tilde{\mathbf{d}}_f = \gamma \odot \Pi_{\mathbb{1}^\perp} \mathbf{d}_f$, where γ is the learned scale and $\Pi_{\mathbb{1}^\perp}$ is the mean-centering projection. The transformation depends only on $\mathbf{d}_f$ and the learned LayerNorm parameters; context enters only through the scalar magnitude $\tilde{\alpha}_f(c)$. Directions remain context-invariant; magnitudes are rescaled by $1/\sigma(\mathbf{h}(c))$. Full proof in Appendix B.

A.6 The Identity-Projection Operator

We now introduce the operator that enables practical applications, and use it to make the framework’s causal claim explicit.

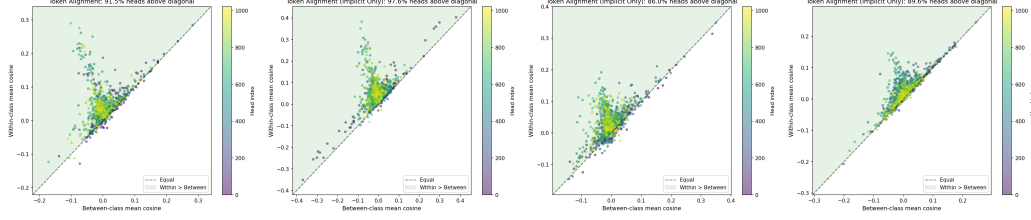


Figure 4: Token alignment validation of the Self-Reference Property across four datasets in LLaMA3-8B. Each point represents one attention head; the x-axis shows mean cosine similarity between class tokens and *other*-class implicit instances, while the y-axis shows similarity to *same*-class implicit instances. Points above the diagonal indicate stronger alignment with the correct class. Percentages indicate heads above diagonal: **Countries** 91.5%, **Animals** 97.6%, **Cartoon Characters** 86.0%, **Emotions** 89.6%. Heads below the diagonal are consistent with head specialization: not every head is expected to encode the class-relevant feature, and some heads may encode shared or opposing subfeatures.

Definition A.23 (Identity-Projection Operator). For a context c and feature f with invariant direction $\mathbf{d}_f$, the *Identity-Projection* is:

$$\mathcal{I}_f(c) \triangleq \mathbf{h}(c)^\top \hat{\mathbf{d}}_f \quad (12)$$

where $\hat{\mathbf{d}}_f = \mathbf{d}_f / \|\mathbf{d}_f\|$ is the unit direction.

The Identity-Projection operator measures the magnitude of feature f within the representation $\mathbf{h}(c)$. By Theorem 3.3, f flows through $\mathbf{h}(c)$ along the fixed direction $\mathbf{d}_f$ with magnitude $\alpha_f(c)$; the projection onto $\mathbf{d}_f$ recovers this magnitude up to the sparse-superposition interference term $\rho(c)$ from Corollary A.15.

Linear-path contribution. The magnitude $\alpha_f(c)$ is not merely a descriptive quantity. Because the architecture reads features through linear interfaces (Definition A.4), $\alpha_f(c)$ is also *the quantity that flows downstream through linear paths*: an interface $L \in \mathcal{L}$ reading the residual stream produces $L\mathbf{h}(c) = \alpha_f(c) L\mathbf{d}_f + L\boldsymbol{\eta}(c)$, so within linear downstream paths, the contribution of f to whatever L computes is proportional to $\alpha_f(c)$. The framework therefore makes a quantitative *linear-path* contribution claim: $\alpha_f(c)$ measures how much f contributes to downstream behavior *when the path from $\mathbf{h}(c)$ to the output is linear*. A feature with $\alpha_f(c) = 0$ is inert through linear paths at c ; a feature with large $\alpha_f(c)$ drives downstream linear computations in proportion. This linear-path account is consistent with the qualitative pattern reported in activation patching studies—intervening on the $\mathbf{d}_f$ -component of $\mathbf{h}(c)$ tends to produce downstream effects scaled by the change in α_f —though we do not run intervention experiments in this paper, so the full causal claim is left to future work.

Proposition A.24 (Feature Operations). For a feature f satisfying the conditions of Theorem 3.3:

- (i) **Detection:** $\mathcal{I}_f(c) > \tau$ indicates f is expressed in c above threshold τ .
- (ii) **Linear-path magnitude:** $\mathcal{I}_f(c)$ measures the magnitude of f in c , which equals (up to scaling by $\|\mathbf{d}_f\|$) the magnitude that flows through any downstream linear interface that reads from $\mathbf{h}(c)$.

Proof. From the decomposition $\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \boldsymbol{\eta}(c)$ in Theorem 3.3:

$$\mathcal{I}_f(c) = \alpha_f(c) \|\mathbf{d}_f\| + \boldsymbol{\eta}(c)^\top \hat{\mathbf{d}}_f. \quad (13)$$

The second term reflects superposition—components of $\boldsymbol{\eta}(c)$ corresponding to other features may have nonzero overlap with $\hat{\mathbf{d}}_f$, since transformer features are not required to occupy orthogonal directions [Elhage et al., 2022]. In practice this overlap is bounded by the sparsity of co-active features and acts as a noise floor on $\mathcal{I}_f$; the central content of (i) and (ii) is that the leading term $\alpha_f(c) \|\mathbf{d}_f\|$ tracks the magnitude of f both as it appears in the representation and as it flows downstream through linear interfaces. $\square$

A.7 Extension to MLP Blocks via Selection-Only Decomposition

Proposition 3.1 applies to features decoded through fixed linear interfaces. A natural concern is whether the framework extends to features that are written by MLP blocks, which contain a per-coordinate nonlinearity. We show that under a structural condition—which we call a *selection-only feature decomposition*—MLP-written features inherit the framework’s invariant-direction conclusions. We then show that 2-layer MLPs structurally satisfy this condition, while deeper MLPs admit a write-direction decomposition but break the independence of feature magnitudes through compositional gating.

Let an MLP block be

$$\text{MLP}(\mathbf{x}) = W_{\text{out}} \cdot \sigma(W_{\text{in}}\mathbf{x} + \mathbf{b}_{\text{in}}) + \mathbf{b}_{\text{out}},$$

with $W_{\text{in}} \in \mathbb{R}^{m \times d}$, $W_{\text{out}} \in \mathbb{R}^{d \times m}$, and σ a per-coordinate nonlinearity. Denote the i -th row of W_{in} by $\mathbf{w}_i^{\text{in}} \in \mathbb{R}^d$ (read direction) and the i -th column of W_{out} by $\mathbf{w}_i^{\text{out}} \in \mathbb{R}^d$ (write direction).

Definition A.25 (Selection-Only Feature Decomposition). An MLP block admits a *selection-only feature decomposition* if there exists a finite collection of feature directions $\{\mathbf{d}_f\}_{f \in F}$ in the residual stream and scalar functions $\{m_f(\mathbf{x})\}_{f \in F}$ such that

$$\text{MLP}(\mathbf{x}) = \sum_{f \in F} m_f(\mathbf{x}) \cdot \mathbf{d}_f + \mathbf{b}_{\text{out}}$$

where:

- (a) **Context-independence of directions.** Each direction $\mathbf{d}_f \in \mathbb{R}^d$ is fixed and does not depend on $\mathbf{x}$. The input controls only the scalar magnitude $m_f(\mathbf{x})$ associated with each direction; it does not change the direction itself.
- (b) **Independent magnitudes.** The contribution along each $\mathbf{d}_f$ is controlled by its own scalar $m_f(\mathbf{x})$, and the value of $m_f(\mathbf{x})$ does not depend on the magnitudes $\{m_{f'}(\mathbf{x})\}_{f' \neq f}$ of other features.

The central claim is condition (a): *directions are context-independent; only magnitudes vary with input*. Whether $m_f(\mathbf{x})$ is large, small, zero, or negative in any given context is not constrained—what is constrained is that the direction $\mathbf{d}_f$ along which f ’s contribution is written remains the same across all contexts. Condition (b) is a structural requirement we use below to distinguish 2-layer from deeper MLPs: each feature’s magnitude is determined independently of other features’ magnitudes, so no feature’s contribution is coupled to the gating of others.

Proposition A.26 (MLP Extension of Invariant Subspace Structure). *Suppose an MLP block admits a selection-only feature decomposition in the sense of Definition A.25. Then for each feature f in the decomposition:*

1. (**Identification**) $\mathbf{d}_f$ is a fixed direction in the residual stream, identifiable from the MLP weights.
2. (**Cross-context invariance**) $\mathbf{d}_f$ does not depend on the input; only $m_f(\mathbf{x})$ does.
3. (**Pathway compatibility**) f is decodable by any downstream linear interface $L \in \mathcal{L}$ whose row space includes $\mathbf{d}_f$, with the decoded value proportional to $m_f(\mathbf{x})$.

The conclusions of Proposition 3.1 thus extend to MLP-written features under the selection-only condition.

Proof sketch. Direct from Definition A.25: (1) and (2) restate condition (a). For (3), any linear functional L applied to $\sum_f m_f(\mathbf{x})\mathbf{d}_f$ yields $\sum_f m_f(\mathbf{x}) \cdot (L\mathbf{d}_f)$, preserving the selection-only structure with modified scalar coefficients. $\square$

Proposition A.27 (Architectural Admissibility of Selection-Only Decomposition). *A 2-layer MLP $\text{MLP}(\mathbf{h}) = W_{\text{out}}\sigma(W_{\text{in}}\mathbf{h} + \mathbf{b}_{\text{in}}) + \mathbf{b}_{\text{out}}$ structurally satisfies the selection-only condition (Definition A.25): the columns of W_{out} are the fixed write directions, and the gates $g_i(\mathbf{h}) = \sigma(\langle \mathbf{w}_i^{\text{in}}, \mathbf{h} \rangle + b_{\text{in},i})$ depend only on each unit’s own read direction. For an L_{out} -layer MLP with $L_{\text{out}} > 2$, the output retains fixed final write directions but the gates depend on compositions of*

upstream activations, breaking the independent-magnitude condition. The framework’s invariant-direction conclusions therefore extend to features written by 2-layer MLP blocks; deeper MLPs preserve write-direction structure but couple feature magnitudes through compositional gating. The proof appears in Appendix B.

Remark A.28 (Empirical Realization of the Dictionary Condition). Whether a trained 2-layer MLP exhibits a selection-only decomposition aligned with semantically meaningful features is an empirical question. The architecture satisfies the structural conditions (Proposition A.27); whether training realizes an interpretable dictionary within this structure requires evidence. Three lines of existing work support this for trained transformers: (i) sparse autoencoder findings showing that MLP activations decompose into sparse, monosemantic features with stable directions [Bricken et al., 2023, Cunningham et al., 2023]; (ii) linear probing across layers showing that class-feature directions remain linearly recoverable through MLP processing [Park et al., 2024]; and (iii) circuit tracing methods such as edge attribution patching, which decompose MLP contributions into per-component additive effects—a decomposition that presupposes selection-only structure. The framework’s structure is therefore: *architecture provides the structural conditions* (Proposition A.27, with the qualitative break at $L > 2$ in condition (b)), while *training provides realization* (cited evidence above).

A.8 The Self-Reference Property

A key practical question is how to obtain a usable reference for $\mathbf{d}_f$ without labeled instance data or direct access to the relevant interface. Class prompts provide such a reference because they are themselves contexts expressing the feature. However, their activations need not be pure: a class prompt may also contain lexical, syntactic, frequency, or category-level components. We therefore state self-reference as an approximate property whose quality depends on the size of the target feature component relative to the residual.

Corollary A.29 (Approximate Self-Reference). *Let t_f be a token or expression that canonically expresses feature f . Assume its activation decomposes as*

$$\mathbf{h}_{t_f} = \beta_f \mathbf{d}_f + \mathbf{u}_f,$$

where $\|\mathbf{d}_f\| = 1$, $\beta_f > 0$, and $\|\mathbf{u}_f\| \leq \delta_f$. If $\delta_f < \beta_f$, then the normalized class-prompt activation

$$\hat{\mathbf{h}}_{t_f} := \frac{\mathbf{h}_{t_f}}{\|\mathbf{h}_{t_f}\|}$$

has positive alignment with the true feature direction. More specifically,

$$\cos(\hat{\mathbf{h}}_{t_f}, \mathbf{d}_f) = \langle \hat{\mathbf{h}}_{t_f}, \mathbf{d}_f \rangle \geq \frac{\beta_f - \delta_f}{\beta_f + \delta_f}.$$

Thus, a class prompt is a good empirical reference for $\mathbf{d}_f$ when its target feature component dominates its residual components.

Proof. We have $\langle \mathbf{h}_{t_f}, \mathbf{d}_f \rangle = \beta_f + \langle \mathbf{u}_f, \mathbf{d}_f \rangle \geq \beta_f - \|\mathbf{u}_f\| \geq \beta_f - \delta_f$. Also, $\|\mathbf{h}_{t_f}\| = \|\beta_f \mathbf{d}_f + \mathbf{u}_f\| \leq \beta_f + \|\mathbf{u}_f\| \leq \beta_f + \delta_f$. Therefore $\cos(\hat{\mathbf{h}}_{t_f}, \mathbf{d}_f) = \langle \mathbf{h}_{t_f}, \mathbf{d}_f \rangle / \|\mathbf{h}_{t_f}\| \geq (\beta_f - \delta_f) / (\beta_f + \delta_f)$. $\square$

Each term in the margin condition maps to a measurable quantity: $\alpha_y(c)$ is the true class signal strength, μ_C is cross-class direction coherence, ρ_j is instance interference along class j ’s direction, and δ_j is class-prompt contamination. Failure modes correspond to violations of specific terms: low α_y (weak signal), high μ_C (overlapping class directions), high ρ_j (instance contains active features along other classes’ directions), or high δ_j (class prompts encode substantial off-feature content).

Remark A.30 (Class prompts as approximate references). The practical implication is that running a class prompt such as “France” through the model can provide a reference direction for the France feature, but only approximately. The class-prompt activation may also encode lexical identity, token frequency, broad category information such as “being a country,” and other contextual components. Mean-centering class prompts, as in Eq. 1, can reduce shared category-level components, while learned unsupervised transformations can further improve separability.

The framework’s name reflects this structure: the model itself, queried with a canonical class prompt, returns an approximate reference for the corresponding feature direction in a single forward pass. The reliability of this reference depends empirically on how strongly the target-feature signal dominates other components of the prompt’s activation.

The mechanism behind Self-Reference: feature aggregation and grouping pressure (full version).

The Self-Reference Property is most natural when explained mechanistically: the same feature dictionary $\{\mathbf{d}_f\}$ that organizes residual activations also organizes how attention writes back to the prediction position and how the unembedding scores candidates. In the regime where attention’s OV map approximately respects the dictionary—a mechanism compatible with the assumption that residual activations across positions are expressed in a shared feature dictionary, which is already implicit in our setup—attention aggregates feature evidence from prompt tokens into the prediction position, where the unembedding then matches it against next-token feature bundles. The following two results formalize this and the gradient-level pressure that reinforces it during training.

Proposition A.31 (Attention Aggregates Shared Feature Evidence). *Consider an attention head writing to prediction position q ,*

$$\mathbf{o}_q = W_O \sum_i a_{q,i} W_V \mathbf{h}_i,$$

with token-position activations decomposing over the shared dictionary $\mathbf{h}_i = \sum_f \alpha_{i,f} \mathbf{d}_f + \boldsymbol{\xi}_i$. Suppose the OV map $M := W_O W_V$ is locally feature-compatible for feature f on the contexts considered, in the sense that

$$M \mathbf{d}_f = \lambda_f \mathbf{d}_f + \mathbf{e}_f, \quad |\langle \mathbf{e}_f, \mathbf{d}_f \rangle| \leq \epsilon_f.$$

We do not require this condition to hold universally across all heads or features; it characterizes the regime in which attention preserves feature f for the head and contexts under analysis. Then the attention-written contribution along $\mathbf{d}_f$ is

$$\langle \mathbf{o}_q, \mathbf{d}_f \rangle = \lambda_f \sum_i a_{q,i} \alpha_{i,f} + E_f(q),$$

where $E_f(q)$ collects cross-feature and residual interference. When many attended tokens express feature f (large $\alpha_{i,f}$), the head’s output along $\mathbf{d}_f$ reflects their attention-weighted aggregate.

Proof. Substitute the decomposition of $\mathbf{h}_i$ into $\mathbf{o}_q$ and use linearity: $\mathbf{o}_q = \sum_i a_{q,i} M \mathbf{h}_i = \sum_i a_{q,i} (\sum_g \alpha_{i,g} (\lambda_g \mathbf{d}_g + \mathbf{e}_g) + M \boldsymbol{\xi}_i)$. Projecting onto $\mathbf{d}_f$ and isolating the $g = f$ matching term gives the stated identity, with $E_f(q)$ collecting the off-feature ($g \neq f$) contributions, the $\mathbf{e}_g$ residuals, and $\langle M \boldsymbol{\xi}_i, \mathbf{d}_f \rangle$. $\square$

Corollary A.32 (Cross-Entropy Reinforces Shared Feature Coordinates). *Let $\alpha_f(q)$ denote the residual-stream magnitude along $\mathbf{d}_f$ at the prediction position, and let unembedding rows decompose as feature bundles $\mathbf{u}_t = \sum_f b_{t,f} \mathbf{d}_f + \mathbf{r}_t$. Up to interference, the logit for token t contains the feature contribution $b_{t,f} \alpha_f(q)$. The cross-entropy gradient with respect to $\alpha_f(q)$ is*

$$\frac{\partial \mathcal{L}}{\partial \alpha_f(q)} = \sum_t p_t(c) b_{t,f} - b_{y,f} + (\text{interference}),$$

which is negative whenever $b_{y,f} > \sum_t p_t(c) b_{t,f}$ —i.e., when the correct next token contains feature f more strongly than the model’s current predicted distribution does. In that regime, gradient descent increases $\alpha_f(q)$.

Proof. The logit $\ell_t(c)$ contains the term $b_{t,f} \alpha_f(q)$ from $\mathbf{u}_t^\top \mathbf{h}(c)$ by Proposition A.1. Differentiating cross-entropy $\mathcal{L} = -\log p_y(c)$ through the softmax gives $\partial \mathcal{L} / \partial \ell_t = p_t(c) - \mathbf{1}[t = y]$, and the chain rule with $\partial \ell_t / \partial \alpha_f(q) = b_{t,f}$ (up to interference) yields the stated identity. $\square$

Together, Proposition A.31 and Corollary A.32 describe a closed loop: attention aggregates feature evidence from prompt tokens into the prediction position; the unembedding scores candidates by feature-bundle matching against this aggregated representation; cross-entropy then reinforces this coordination during training, increasing the shared feature magnitude whenever the target token contains the feature more strongly than the prediction does. Repeated across many contexts, this is a *grouping pressure*: surface forms that all express the same underlying feature are routed through the same output-facing coordinate because they all support next tokens whose unembedding bundles include the corresponding feature component. We do not claim this proves features must emerge—emergence still depends on optimization dynamics, training data, and inductive biases—but it provides the mechanistic content of *why* shared feature directions are useful solutions to next-token prediction once the architecture commits to linear or selection-only readouts. The connection to next-token prediction as implicit classification is made explicit in Remark A.21.

1040 A.9 Empirical Validation of the Self-Reference Property

1041 Corollary A.29 predicts that the activation $\mathbf{h}_t$ produced by a class prompt t should align with the
 1042 activations $\mathbf{h}(c)$ produced by other contexts c expressing the same feature. We test this by comparing
 1043 each class prompt’s hidden state against the mean instance hidden state for each class, computed
 1044 across attention heads.

1045 Specifically, for each attention head and class k , we compute the cosine similarity between the class
 1046 token direction $\mathbf{h}_{t_k}$ and the mean instance direction $\bar{\mathbf{h}}_k$ for the same class (within-class) versus other
 1047 classes (between-class). If tokens provide feature directions as predicted, within-class similarity
 1048 should consistently exceed between-class similarity.

1049 Figure 4 confirms this prediction: across 89.6%-98.6% of attention heads, the class token aligns
 1050 more strongly with its own class instances than with any other class. This pattern validates the
 1051 Self-Reference Property—tokens encode the same directional features present in contexts expressing
 1052 those features, enabling zero-shot extraction of semantic directions.

1053 The 2–14% of heads that fall below the diagonal are predicted by the framework rather than coun-
 1054 terexamples to it. Each attention head encodes a particular subset of features (positional, syntactic,
 1055 copying, semantic), and a class token aligns only with heads whose features it actually uses; heads
 1056 that encode unrelated features show no alignment. A second source of off-diagonal points is *partial-*
 1057 *synonym* structure: a feature shared by multiple classes (e.g., a feature common to “cat” and “dog”)
 1058 will, in a head that encodes it, align both classes to the same direction. Instances remain more aligned
 1059 with their own class overall, because the larger set of class-relevant heads—in both number and
 1060 magnitude—dominates the aggregate, even when no individual head encodes the whole class.

1061 B Proofs of Main Results

1062 B.1 Proofs of Lemma A.1, Propositions 3.1 and A.1, and Theorem 3.3

1063 Proof of Lemma A.1 (Linear-Path Communication Lemma).

1064 *Proof.* Let $L \in \mathcal{L}$ be a fixed linear interface and $\phi \in \mathbb{R}^{k_L}$ a fixed downstream readout direction. The
 1065 downstream scalar quantity is

$$z(c) = \phi^\top L \mathbf{h}(c) = (L^\top \phi)^\top \mathbf{h}(c) = \mathbf{w}^\top \mathbf{h}(c), \quad (14)$$

1066 where $\mathbf{w} := L^\top \phi \in \mathbb{R}^d$. The direction $\mathbf{w}$ is fully determined by L (a fixed model parameter, e.g.,
 1067 W_U or an attention QKV matrix) and ϕ (a fixed readout direction in interface space, e.g., the indicator
 1068 vector $\mathbf{e}_t$ for a token t). Since both are fixed at inference, $\mathbf{w}$ does not depend on c . No empirical
 1069 estimation or supervised training is required to identify $\mathbf{w}$. $\square$

1070 Proof of Proposition 3.1 (Linear Unembedding Imposes Fixed Contrast Coordinates).

1071 *Proof.* The full proof is given in the main text. We restate it briefly. The inference identity $\ell_y(c) -$
 1072 $\ell_j(c) = (\mathbf{u}_y - \mathbf{u}_j)^\top \mathbf{h}^{(L)}(c)$ follows directly from the linearity of the unembedding, since $\ell(c) =$
 1073 $W_U \mathbf{h}^{(L)}(c)$ and $\mathbf{u}_t$ is the t -th row of W_U .

1074 For the training-time identity, let $\ell(c) = W_U \mathbf{h}^{(L)}(c)$ and $\mathbf{p}(c) = \text{softmax}(\ell(c))$. The cross-entropy
 1075 loss for target token y is $\mathcal{L}(c) = -\ell_y(c) + \log \sum_t \exp(\ell_t(c))$. Differentiating with respect to logits
 1076 gives $\partial \mathcal{L} / \partial \ell_t = p_t(c) - \mathbf{1}[t = y]$, so $\nabla_{\ell} \mathcal{L} = \mathbf{p}(c) - \mathbf{e}_y$. The chain rule with $\ell = W_U \mathbf{h}^{(L)}$ then
 1077 gives

$$\nabla_{\mathbf{h}^{(L)}} \mathcal{L}(c) = W_U^\top (\mathbf{p}(c) - \mathbf{e}_y) = \sum_{t \in \mathcal{V}} (p_t(c) - \mathbf{1}[t = y]) \mathbf{u}_t. \quad (15)$$

1078 Since $\sum_t (p_t(c) - \mathbf{1}[t = y]) = 0$, the gradient rearranges as $\sum_{j \neq y} p_j(c) (\mathbf{u}_j - \mathbf{u}_y)$, a probability-
 1079 weighted sum of contrast directions $\mathbf{u}_j - \mathbf{u}_y$. The same fixed contrast directions that govern inference
 1080 therefore govern training-time updates as well. $\square$

1081 **Proof of Proposition A.1 (Nonlinear Readouts Relax Output-Facing Directional Pressure).**

1082 *Proof.* For a general readout $\ell(c) = G(\mathbf{h}^{(L)}(c))$, the chain rule gives $\nabla_{\mathbf{h}^{(L)}} \mathcal{L} = J_G(\mathbf{h}^{(L)})^\top (\mathbf{p}(c) -$
 1083 $\mathbf{e}_y)$. The Jacobian rows define the directions through which the loss applies pressure to $\mathbf{h}^{(L)}$; when
 1084 G is nonlinear, these depend on the activation and hence on c .

1085 For the 2-layer MLP $G(\mathbf{h}) = W_{\text{out}} \sigma(W_{\text{in}} \mathbf{h} + \mathbf{b})$, differentiating layer by layer:

$$J_G(\mathbf{h}) = W_{\text{out}} \text{diag}(\sigma'(W_{\text{in}} \mathbf{h} + \mathbf{b})) W_{\text{in}}. \quad (16)$$

1086 Therefore

$$\nabla_{\mathbf{h}^{(L)}} \mathcal{L} = W_{\text{in}}^\top \text{diag}(\sigma'(\cdot)) W_{\text{out}}^\top (\mathbf{p}(c) - \mathbf{e}_y), \quad (17)$$

1087 which is a sum over the columns of $W_{\text{in}}^\top$ (the read directions of the MLP head), with weights
 1088 determined by the per-context gates $\sigma'(\cdot)$ and the prediction-error vector. The columns of $W_{\text{in}}^\top$ are
 1089 fixed; only their scalar weights depend on c . Forward writes through the head use the fixed columns
 1090 of W_{out} ; backward pressure on $\mathbf{h}^{(L)}$ uses the fixed columns of $W_{\text{in}}^\top$. Selection-only structure is
 1091 preserved in both directions. $\square$

1092 **Proof of Theorem 3.3 (Invariant Direction under Sparse Superposition).**

1093 *Proof.* The full proof is given in the main text following the theorem statement. We restate the
 1094 steps for completeness. Let $\mathcal{D} = \{\mathbf{d}_g\}_{g \in \mathcal{F}}$ be a unit-norm dictionary with coherence μ , and assume
 1095 $\mathbf{h}(c) = \sum_{g \in S(c)} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c)$ with $|S(c)| \leq k$ and $\|\boldsymbol{\xi}(c)\| \leq \varepsilon$.

1096 Separating the f term (taking $\alpha_f(c) = 0$ if $f \notin S(c)$),

$$\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \sum_{g \in S(c), g \neq f} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c). \quad (18)$$

1097 Projecting onto $\mathbf{d}_f$ with $\|\mathbf{d}_f\| = 1$,

$$\langle \mathbf{h}(c), \mathbf{d}_f \rangle = \alpha_f(c) + \sum_{g \in S(c), g \neq f} \alpha_g(c) \langle \mathbf{d}_g, \mathbf{d}_f \rangle + \langle \boldsymbol{\xi}(c), \mathbf{d}_f \rangle. \quad (19)$$

1098 By the triangle inequality, the coherence bound $|\langle \mathbf{d}_g, \mathbf{d}_f \rangle| \leq \mu$, and Cauchy–Schwarz with $\|\mathbf{d}_f\| = 1$
 1099 and $\|\boldsymbol{\xi}(c)\| \leq \varepsilon$,

$$|\langle \mathbf{h}(c), \mathbf{d}_f \rangle - \alpha_f(c)| \leq \mu \sum_{g \in S(c), g \neq f} |\alpha_g(c)| + \varepsilon. \quad (20)$$

1100 The simplified bound $\mu(k-1)A + \varepsilon$ follows from $|S(c) \setminus \{f\}| \leq k-1$ and the assumption
 1101 $|\alpha_g(c)| \leq A$ for all active non- f features. $\square$

1102 **B.2 Proof of Proposition C.1 (Dimensional Bound)**

1103 *Proof.* By Proposition 3.1, there exists a context-invariant subspace $\mathcal{S}_f$ containing all f -relevant
 1104 information. We show that linear separability of K feature values constrains $\dim(\mathcal{S}_f) \leq K-1$.

1105 Let $\{\mathbf{h}_{y_k}\}_{k=1}^K \subset \mathcal{S}_f$ be representative embeddings (e.g., class centroids) for each feature value y_k .
 1106 For these to be linearly separable, they must be affinely independent: no $\mathbf{h}_{y_k}$ lies in the affine hull of
 1107 the others.

1108 The affine hull of K points has dimension at most $K-1$. To see this, define centered representations:

$$\tilde{\mathbf{h}}_{y_k} = \mathbf{h}_{y_k} - \bar{\mathbf{h}}, \quad \text{where } \bar{\mathbf{h}} = \frac{1}{K} \sum_{k=1}^K \mathbf{h}_{y_k} \quad (21)$$

1109 These centered vectors satisfy $\sum_k \tilde{\mathbf{h}}_{y_k} = \mathbf{0}$, so at most $K-1$ are linearly independent. Thus
 1110 $\text{span}(\{\tilde{\mathbf{h}}_{y_k}\}) \leq K-1$.

1111 Since linear separability requires the f -relevant subspace to distinguish all K values, and this can be
 1112 achieved in $K-1$ dimensions, the minimal invariant subspace satisfies $\dim(\mathcal{S}_f) \leq K-1$.

1113 For binary features ($K=2$), this bound yields $\dim(\mathcal{S}_f) \leq 1$. Combined with the requirement that
 1114 $\mathcal{S}_f$ be non-trivial (the feature must be detectable), we have $\dim(\mathcal{S}_f) = 1$, establishing directional
 1115 invariance. $\square$

1116 B.3 Proof of Proposition A.27 (Architectural Admissibility)

1117 *Proof. 2-layer case.* For a 2-layer MLP $\text{MLP}(\mathbf{x}) = W_{\text{out}} \cdot \sigma(W_{\text{in}}\mathbf{x} + \mathbf{b}_{\text{in}}) + \mathbf{b}_{\text{out}}$, expand the
 1118 matrix product directly:

$$\text{MLP}(\mathbf{x}) - \mathbf{b}_{\text{out}} = W_{\text{out}} \cdot \sigma(W_{\text{in}}\mathbf{x} + \mathbf{b}_{\text{in}}) = \sum_{i=1}^m \sigma(\langle \mathbf{w}_i^{\text{in}}, \mathbf{x} \rangle + b_{\text{in},i}) \cdot \mathbf{w}_i^{\text{out}}. \quad (22)$$

1119 Setting $g_i(\mathbf{x}) := \sigma(\langle \mathbf{w}_i^{\text{in}}, \mathbf{x} \rangle + b_{\text{in},i})$, we have $\text{MLP}(\mathbf{x}) = \sum_i g_i(\mathbf{x}) \mathbf{w}_i^{\text{out}} + \mathbf{b}_{\text{out}}$, identifying
 1120 $\mathbf{d}_i = \mathbf{w}_i^{\text{out}}$ and $m_i(\mathbf{x}) = g_i(\mathbf{x})$.

1121 *Verification of Definition A.25.* (a) Context-independence: each write direction $\mathbf{w}_i^{\text{out}}$ is a column
 1122 of W_{out} , a fixed model parameter, so $\mathbf{d}_i$ does not depend on $\mathbf{x}$. (b) Independent magnitudes: each
 1123 gate $g_i(\mathbf{x})$ is computed by applying σ to a single linear projection $\langle \mathbf{w}_i^{\text{in}}, \mathbf{x} \rangle$. Because σ is applied per
 1124 coordinate, the value of $g_j(\mathbf{x})$ for $j \neq i$ does not enter the computation of $g_i(\mathbf{x})$, so the magnitudes
 1125 are determined independently across features.

1126 **Deeper case ($L > 2$).** For an L -layer MLP with weights $W_1, \dots, W_L$ and biases $\mathbf{b}_1, \dots, \mathbf{b}_L$:

$$\text{MLP}_L(\mathbf{x}) = W_L \cdot \sigma(W_{L-1} \cdot \sigma(\dots \sigma(W_1\mathbf{x} + \mathbf{b}_1) \dots) + \mathbf{b}_{L-1}) + \mathbf{b}_L. \quad (23)$$

1127 Write this as $\text{MLP}_L(\mathbf{x}) = W_L \cdot \mathbf{v}(\mathbf{x}) + \mathbf{b}_L$. Suppose, for contradiction, that this admits a selection-
 1128 only decomposition with directions $\{\mathbf{d}_f\}$ taken from columns of W_L and scalars $\{m_f(\mathbf{x})\}$. Then
 1129 condition (b) requires that the contribution along each column $\mathbf{d}_i$ depends on $\mathbf{x}$ only through a scalar
 1130 $m_i(\mathbf{x})$ that is independent of the other gates' values.

1131 But the i -th coordinate of $\mathbf{v}(\mathbf{x})$ is:

$$v_i(\mathbf{x}) = \sigma\left(\sum_j (W_{L-1})_{ij} \cdot u_j(\mathbf{x}) + (\mathbf{b}_{L-1})_i\right) \quad (24)$$

1132 where each $u_j(\mathbf{x})$ is itself the post-nonlinearity activation of an upstream layer, and depends on
 1133 multiple coordinates of yet-earlier layers' outputs. The scalar $v_i(\mathbf{x})$ is therefore a composite quantity
 1134 that mixes the values of multiple intermediate gates: changing the value of $u_j(\mathbf{x})$ for $j \neq i$ generally
 1135 changes $v_i(\mathbf{x})$ via the linear combination inside σ . Hence the magnitude associated with column i of
 1136 W_L is coupled to the values of other intermediate activations and cannot be determined independently
 1137 of other features' magnitudes. Condition (b) of Definition A.25 therefore fails. The same conclusion
 1138 holds for any other choice of $\{\mathbf{d}_f\}$ inside the column span of W_L , since the coupling is intrinsic to
 1139 the composition of multiple nonlinearities. $\square$

1140 C Capacity Constraints and Feature Factorization

1141 C.1 Detailed Argument for Capacity-Driven Feature Sharing

1142 We provide a complete argument that capacity constraints in transformers favor factorized representa-
 1143 tions with shared, invariant feature directions.

1144 *Proposition* (Capacity Constraint Implies Feature Sharing, restated). Let $\mathcal{M}$ be a transformer with
 1145 vocabulary $|\mathcal{V}|$ and hidden dimension d , where $|\mathcal{V}| \gg d$. Under the following conditions:

- 1146 1. **Linear readout:** Token logits are computed via $\text{logit}_t = \mathbf{w}_t^\top \mathbf{h}(c)$
- 1147 2. **Sparse activation:** Each context c expresses a small subset of all possible features
- 1148 3. **Shared features:** Multiple tokens share semantic attributes

1149 Then the optimal representation factorizes tokens as:

$$\mathbf{w}_t = \sum_{f \in F_t} \alpha_{t,f} \mathbf{d}_f \quad (25)$$

1150 where F_t is the feature set for token t , and $\{\mathbf{d}_f\}$ are shared feature directions with $|F| \ll |\mathcal{V}|$.

1151 *Proof.* We proceed in four steps, establishing that factorization is optimal under capacity constraints
 1152 when features are sparse.

1153 **Step 1: Dimensional constraint.** The hidden dimension d bounds the number of orthogonal
 1154 directions available. With typical values $|\mathcal{V}| \approx 50,000$ and $d \approx 4,000$, at most d tokens can have
 1155 mutually orthogonal representations in W_U . The remaining $|\mathcal{V}| - d$ tokens must share directions with
 1156 others, which creates potential interference in the readout.

1157 **Step 2: Interference cost depends on co-occurrence.** Suppose tokens t_1 and t_2 share a direction
 1158 $\mathbf{d}$, i.e., both $\mathbf{w}_{t_1}$ and $\mathbf{w}_{t_2}$ have non-zero projections onto $\mathbf{d}$. Interference occurs when both tokens are
 1159 relevant predictions for a context c : the hidden state $\mathbf{h}(c)$ cannot independently modulate the logits
 1160 for t_1 and t_2 along their shared direction. The expected interference cost is:

$$\mathcal{L}_{\text{interference}}(t_1, t_2) \propto P(t_1, t_2 \text{ both relevant in } c) \cdot |\mathbf{w}_{t_1}^\top \mathbf{d}| \cdot |\mathbf{w}_{t_2}^\top \mathbf{d}| \quad (26)$$

1161 For sparse features—those with low co-occurrence probability across the corpus—this interference
 1162 cost is small.

1163 **Step 3: Factorization minimizes total representation cost.** Consider the total cost of a represen-
 1164 tation as the sum of dimensional cost (number of directions used) and interference cost. Compare
 1165 two strategies:

- 1166 • *Unique directions:* Each token t receives a dedicated direction $\mathbf{d}_t$. This eliminates interfer-
 1167 ence but requires $|\mathcal{V}|$ dimensions—infeasible when $|\mathcal{V}| \gg d$.
- 1168 • *Factorized features:* Tokens share $|F|$ feature directions, where each token t is represented as
 1169 a sparse combination of features $f \in F_t$. This requires only $|F|$ dimensions plus interference
 1170 costs proportional to feature co-occurrence.

1171 When semantic features are sparse and $|F| \ll |\mathcal{V}|$, the factorized strategy achieves lower total cost.
 1172 Intuitively, “mammal” and “European” rarely co-occur as the dominant semantic features of a context,
 1173 so tokens sharing these directions (e.g., “cat” and “France”) experience minimal interference despite
 1174 their shared structure.

1175 **Step 4: Each factor is communicable and invariant.** Under factorization, the logit computation
 1176 for token t decomposes as:

$$\text{logit}_t = \mathbf{w}_t^\top \mathbf{h}(c) = \sum_{f \in F_t} \alpha_{t,f} \mathbf{d}_f^\top \mathbf{h}(c) = \sum_{f \in F_t} \alpha_{t,f} \phi_f(c) \quad (27)$$

1177 where $\phi_f(c) = \mathbf{d}_f^\top \mathbf{h}(c)$ is the activation of feature f in context c . For the model to correctly predict
 1178 token probabilities, the hidden state $\mathbf{h}(c)$ must encode each relevant feature f such that $\phi_f(c)$ reflects
 1179 the appropriate magnitude. This imposes three requirements on each feature direction $\mathbf{d}_f$:

- 1180 1. **Linearly decodable:** The feature activation is extracted via a linear projection $\mathbf{d}_f^\top \mathbf{h}(c)$.
- 1181 2. **Multi-context:** The same direction $\mathbf{d}_f$ is used across all tokens t with $f \in F_t$ and all
 1182 contexts c where feature f is relevant.
- 1183 3. **Context-invariant:** The direction $\mathbf{d}_f$ is fixed in W_U and does not depend on c ; only the
 1184 magnitude $\phi_f(c)$ varies with context.

1185 These three properties—linear decodability, multi-context usage, and context-invariance—are pre-
 1186 cisely the conditions of Theorem 3.3. Therefore, each semantic factor $\mathbf{d}_f$ occupies an invariant
 1187 subspace in the representation geometry. $\square$

1188 **Connection to prior work.** This result connects to the superposition hypothesis [Elhage et al.,
 1189 2022], which demonstrates that neural networks exploit high-dimensional geometry to represent more
 1190 features than dimensions when features are sparse. Our contribution is to show that this superposition,
 1191 combined with the linear readout constraint of transformers, shapes the specific geometric form
 1192 of feature representations: fixed directions with context-dependent magnitudes. While Achille
 1193 and Soatto [2018] establish that information-minimal representations are necessarily invariant, we
 1194 characterize the *form* this invariance takes within transformer architectures.

1195 C.2 Proof of Proposition A.22

1196 *Proof.* We analyze each component of Layer Normalization.

1197 **Step 1: Mean-centering.** Define the mean-centering projection

$$\Pi_{\mathbb{1}^\perp} := I - \frac{1}{d} \mathbb{1} \mathbb{1}^\top,$$

1198 where $\mathbb{1} \in \mathbb{R}^d$ is the all-ones vector. Applying this linear operator to our decomposition:

$$\Pi_{\mathbb{1}^\perp} h(c) = \alpha_f(c) \Pi_{\mathbb{1}^\perp} d_f + \Pi_{\mathbb{1}^\perp} \eta(c).$$

1199 Since $d_f \notin \text{span}(\mathbb{1})$ by assumption, the projected direction $d'_f := \Pi_{\mathbb{1}^\perp} d_f \neq 0$.

1200 **Step 2: Scalar normalization.** LayerNorm divides by the standard deviation $\sigma(h(c)) > 0$:

$$\frac{\Pi_{\mathbb{1}^\perp} h(c)}{\sigma(h(c))} = \frac{\alpha_f(c)}{\sigma(h(c))} d'_f + \frac{\Pi_{\mathbb{1}^\perp} \eta(c)}{\sigma(h(c))}.$$

1201 Since $\sigma(h(c))$ is a positive scalar, this operation preserves the subspace $\text{span}(d'_f)$.

1202 **Step 3: Learned affine transformation.** LayerNorm applies elementwise scaling by $\gamma \in \mathbb{R}^d$ and
1203 adds bias $\beta \in \mathbb{R}^d$:

$$\text{LN}(h(c)) = \frac{\alpha_f(c)}{\sigma(h(c))} (\gamma \odot d'_f) + \frac{\gamma \odot \Pi_{\mathbb{1}^\perp} \eta(c)}{\sigma(h(c))} + \beta.$$

1204 Define the transformed feature direction

$$\tilde{d}_f := \gamma \odot d'_f = \gamma \odot \Pi_{\mathbb{1}^\perp} d_f.$$

1205 This direction depends only on d_f and the learned parameter γ , and is therefore context-invariant.

1206 LayerNorm maps the context-invariant subspace $\text{span}(d_f)$ to a new context-invariant subspace
1207 $\text{span}(\tilde{d}_f)$ via the fixed transformation $d_f \mapsto \gamma \odot \Pi_{\mathbb{1}^\perp} d_f$. The feature remains affinely decodable:
1208 a linear probe along $\tilde{d}_f$ with appropriate bias recovers a signal proportional to $\alpha_f(c)$, scaled by
1209 $1/\sigma(h(c))$. $\square$

1210 C.3 Proof of Proposition C.1 (Dimensional Bound)

1211 **Proposition C.1** (Dimensional Bound). *If f takes K linearly separable values, then $\dim(\mathcal{S}_f) \leq$*
1212 *$K - 1$. For binary features, this yields directional invariance: $\mathcal{S}_f = \text{span}(\{\mathbf{d}_f\})$.*

1213 C.4 Proof of Corollary A.29 (Approximate Self-Reference)

1214 *Corollary* (Approximate Self-Reference, restated). Let t_f be a token or expression that canonically
1215 expresses feature f . Assume its activation decomposes as $\mathbf{h}_{t_f} = \beta_f \mathbf{d}_f + \mathbf{u}_f$, where $\|\mathbf{d}_f\| = 1$,
1216 $\beta_f > 0$, and $\|\mathbf{u}_f\| \leq \delta_f < \beta_f$. Then the normalized class-prompt activation $\hat{\mathbf{h}}_{t_f} := \mathbf{h}_{t_f} / \|\mathbf{h}_{t_f}\|$
1217 satisfies

$$\cos(\hat{\mathbf{h}}_{t_f}, \mathbf{d}_f) \geq \frac{\beta_f - \delta_f}{\beta_f + \delta_f}.$$

1218 *Proof.* By Definition A.2, the activation $\mathbf{h}_{t_f}$ is the hidden state at some layer and position obtained by
1219 running the token (or expression) t_f through the model; it is an element of the context set $\mathcal{C}$. Because
1220 t_f canonically expresses f , applying Assumption A.12 to this context, the activation admits a sparse
1221 decomposition with $f \in S(t_f)$ and $\alpha_f(t_f) \neq 0$. Identifying $\beta_f := \alpha_f(t_f)$ (assumed positive without
1222 loss of generality, since the sign of $\mathbf{d}_f$ is a convention) and $\mathbf{u}_f := \sum_{g \in S(t_f), g \neq f} \alpha_g(t_f) \mathbf{d}_g + \boldsymbol{\xi}(t_f)$ as
1223 the residual contribution from non- f active features and noise, the hypothesis $\|\mathbf{u}_f\| \leq \delta_f$ corresponds
1224 to a bound on the magnitude of these other components in the class prompt's activation.

1225 For the numerator,

$$\langle \mathbf{h}_{t_f}, \mathbf{d}_f \rangle = \beta_f + \langle \mathbf{u}_f, \mathbf{d}_f \rangle \geq \beta_f - \|\mathbf{u}_f\| \|\mathbf{d}_f\| \geq \beta_f - \delta_f, \quad (28)$$

1226 using Cauchy–Schwarz and $\|\mathbf{d}_f\| = 1$.

1227 For the denominator,

$$\|\mathbf{h}_{t_f}\| = \|\beta_f \mathbf{d}_f + \mathbf{u}_f\| \leq \beta_f + \|\mathbf{u}_f\| \leq \beta_f + \delta_f \quad (29)$$

1228 by the triangle inequality. Combining,

$$\cos(\widehat{\mathbf{h}}_{t_f}, \mathbf{d}_f) = \frac{\langle \mathbf{h}_{t_f}, \mathbf{d}_f \rangle}{\|\mathbf{h}_{t_f}\|} \geq \frac{\beta_f - \delta_f}{\beta_f + \delta_f}. \quad (30)$$

1229 The bound is positive whenever $\delta_f < \beta_f$, which is the condition that the target-feature component
 1230 dominates the residual. Note that this proof does not single out any specific interface or layer: it
 1231 applies wherever Assumption A.12 applies, so class-prompt activations can be extracted from any
 1232 layer or attention head and used as approximate references; the choice of extraction point is an
 1233 empirical matter. $\square$

1234 C.5 Additional Theoretical Implications

1235 Our framework yields several additional consequences for transformer interpretability:

1236 **Corollary C.2** (Attention Preserves Feature Identity). *The OV circuit linearly transforms features*
 1237 *while preserving their identity. If position j encodes feature f as $\mathbf{h}_j = \alpha_f \mathbf{d}_f + \boldsymbol{\eta}_j$, the attention*
 1238 *output at position i is:*

$$\mathbf{o}_i^{(f)} = \underbrace{\left(\sum_j a_{ij} \cdot \alpha_f^{(j)} \right)}_{\text{context-dependent magnitude}} \cdot \underbrace{W_O W_V \mathbf{d}_f}_{\text{linearly transformed direction}} \quad (31)$$

1239 *The QK circuit modulates how much of the feature transfers (magnitude); the OV circuit determines*
 1240 *how the feature direction is transformed.*

1241 *Proof.* By linearity of the OV circuit:

$$\mathbf{o}_i = \sum_j a_{ij} \cdot W_O W_V \mathbf{h}_j \quad (32)$$

$$= \sum_j a_{ij} \cdot W_O W_V (\alpha_f^{(j)} \mathbf{d}_f + \boldsymbol{\eta}_j) \quad (33)$$

$$= \left(\sum_j a_{ij} \cdot \alpha_f^{(j)} \right) W_O W_V \mathbf{d}_f + \sum_j a_{ij} \cdot W_O W_V \boldsymbol{\eta}_j \quad (34)$$

1242 The f -relevant component is the first term, with magnitude determined by attention-weighted sum
 1243 and direction determined by linear transformation of $\mathbf{d}_f$. $\square$

1244 **Corollary C.3** (Distributional Influence). *Adding a feature direction $\mathbf{d}_f$ to the hidden state shifts the*
 1245 *output distribution predictably:*

$$\Delta \text{logit}_t = \lambda \cdot \mathbf{w}_t^\top \mathbf{d}_f \quad (35)$$

1246 *Tokens aligned with $\mathbf{d}_f$ are boosted; anti-aligned tokens are suppressed. This provides the theoretical*
 1247 *foundation for activation steering [Turner et al., 2023, Zou et al., 2023].*

1248 *Proof.* Let $\mathbf{h}$ be the original hidden state and $\mathbf{h}' = \mathbf{h} + \lambda \mathbf{d}_f$ the steered state. By linearity of
 1249 unembedding:

$$\text{logit}_t(\mathbf{h}') = \mathbf{w}_t^\top \mathbf{h}' = \mathbf{w}_t^\top \mathbf{h} + \lambda \mathbf{w}_t^\top \mathbf{d}_f = \text{logit}_t(\mathbf{h}) + \lambda \mathbf{w}_t^\top \mathbf{d}_f$$

1250 Thus $\Delta \text{logit}_t = \lambda \mathbf{w}_t^\top \mathbf{d}_f$, which is positive when $\mathbf{w}_t$ and $\mathbf{d}_f$ are aligned. $\square$

1251 **Corollary C.4** (Non-Bidirectionality). *The invariant subspace for “A predicts B” is generally*
 1252 *distinct from that for “B predicts A”:*

$$\mathcal{S}_{A \rightarrow B} \not\subseteq \mathcal{S}_{B \rightarrow A} \quad \text{and} \quad \mathcal{S}_{B \rightarrow A} \not\subseteq \mathcal{S}_{A \rightarrow B} \quad (36)$$

1253 *This provides a geometric explanation for the reversal curse [Berglund et al., 2023].*

1254 *Proof.* The features “ A predicts B ” and “ B predicts A ” are distinct communicable features—they
 1255 produce different outputs and are learned from different training examples. By Proposition 3.1, each
 1256 occupies its own invariant subspace determined by that feature. Since these are different features,
 1257 their subspaces need not coincide. Models trained on “ A is B ” learn $\mathcal{S}_{A \rightarrow B}$ but have no reason to
 1258 simultaneously learn $\mathcal{S}_{B \rightarrow A}$, explaining the failure to infer reverse relations. $\square$

1259 D Additional Testing on Unsupervised Probes

1260 D.1 Unsupervised Probes

1261 We validate these properties through comprehensive visualization analysis, comparing embeddings
 1262 before and after probe transformation using both PCA and t-SNE projections. These visualizations
 1263 confirm that unsupervised probes learn transformations that improve class separability—instances
 1264 cluster more tightly around their corresponding prototypes in the transformed space—while the
 1265 prototype vectors themselves maintain interpretable directional structure aligned with semantic
 1266 features.

1267 Zero-shot generalization to unseen classes. PCA (top) and t-SNE (bottom) projections of month
 1268 embeddings after unsupervised probe transformation. Training used only odd months; even months
 1269 were held out. In both visualizations, held-out classes cluster tightly around their respective prototypes,
 1270 demonstrating that the learned geometric structure generalizes without exposure to these classes
 1271 during training.

1272 E Additional Experiments

1273 This section reports additional experiments that validate the robustness, statistical significance, and
 1274 scope of the framework on more complex tasks and across different computational pathways.

1275 E.1 Polysemy as the inverse of *Partial Synonyms*

1276 Polysemy provides a natural test case for the directional account: “Apple” encodes both fruit and
 1277 company meanings as superposed directions. Table 2 shows both Fruits and Companies datasets
 1278 classify accurately despite sharing this token. To test explicit disentanglement, we combine both
 1279 datasets and compare using a single “Apple” token versus distinct tokens (“Fruit apple”, “Company
 1280 Apple”). Both strategies achieve comparable accuracy (Fig. 7: 69% vs. 65.7%), confirming that both
 1281 meanings co-exist in superposition—context modulates magnitude, not selection.

1282 This geometric view inverts naturally: “Dog” and “Cat” are distinct tokens sharing a component along
 1283 $\mathbf{d}_{\text{mammal}}$ —*partial synonyms* with respect to that feature. Polysemy (one token, multiple directions)
 1284 and partial synonymy (multiple tokens, shared direction) are two sides of the same coin.

1285 E.2 Supervised Linear Probe Baseline

1286 The methods compared in Table 2 are intentionally restricted to those that do not use labeled instance
 1287 data, since the framework targets unsupervised feature recovery via class prompts. To calibrate how
 1288 much performance is recoverable in the fully supervised regime, we report regularized supervised
 1289 linear probes as an upper bound. For each (task, model) cell, we train an L2-regularized logistic
 1290 regression classifier on attention-head activations using a 70/15/15 train/validation/test split, with the
 1291 regularization strength selected on the validation set. We report held-out test accuracy.

1292 The relevant calibration question is not whether class-prompt methods match supervised probes (they
 1293 typically will not, since the supervised probe has direct access to class labels), but rather *how much of*
 1294 *the supervised linear accessibility is recovered by class-prompt methods alone*. The answer informs
 1295 the practical scope of the Self-Reference Property: a small gap indicates that class-prompt activations
 1296 carry most of the linearly accessible feature information, while a large gap indicates that supervision
 1297 provides additional separability beyond what class prompts encode. The numerical comparison and
 1298 per-task gap analysis will be released alongside the final version, pending the additional supervised
 1299 runs.

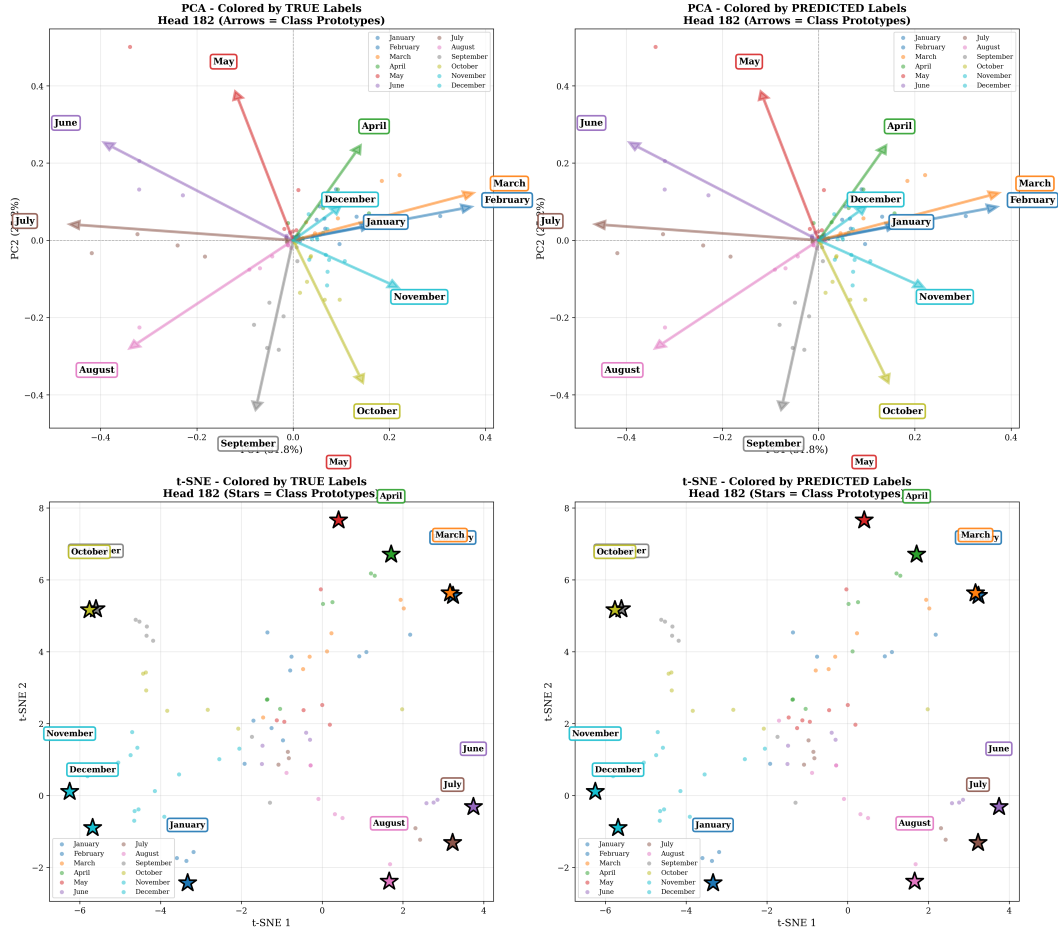


Figure 5: Zero-shot generalization of unsupervised probes. Probes trained only on odd months (Jan, Mar, May, Jul, Sep, Nov) successfully classify held-out even months, demonstrating that the learned transformation captures generalizable temporal structure rather than memorizing training classes.

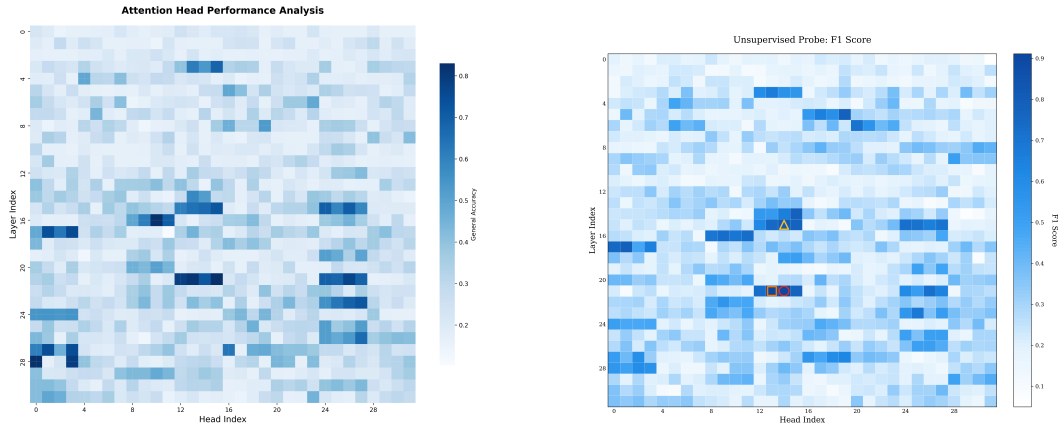


Figure 6: Zero-shot (left) and Unsupervised prob classification performance on the Animal dataset in Llama-3-8B. Both contain similar accuracy distributions.

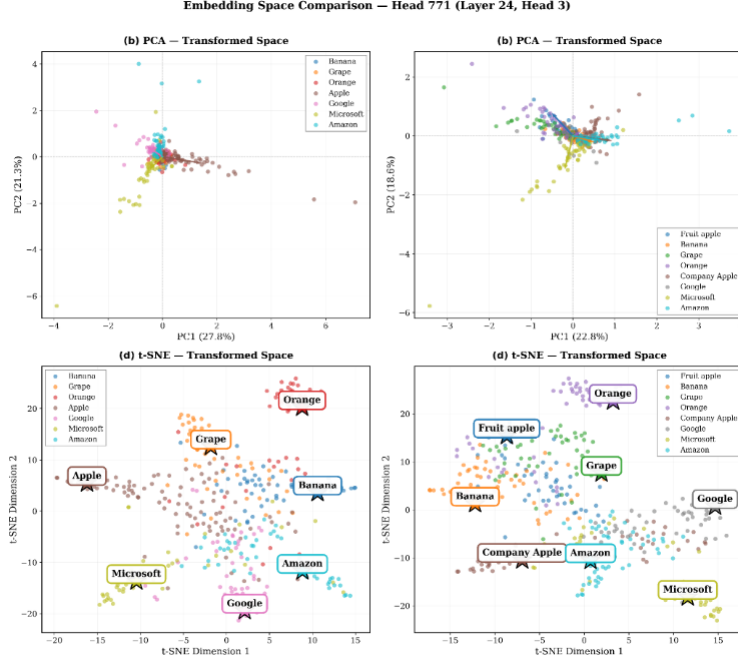


Figure 7: PCA and t-SNE projections of embeddings for the polysemous word Apple (fruit vs. company) using domain-specific tokens. (Left) A single “Apple” token representing both classes (69% accuracy). (Right) Separate tokens “Fruit apple” and “Company Apple” treated as distinct classes (65.7% accuracy). Both methods successfully disentangle the two senses, with instances clustering around their respective class prototypes.

E.3 Robustness to Input Perturbations

We evaluate stability under three types of perturbations: (1) syntax-level shuffling, (2) random-noise injection, and (3) adversarial system-level prefixes. As shown in Table 6, all methods behave similarly under syntax and noise perturbations, indicating robustness to surface-level variations. Under adversarial prefixes, however, a difference emerges: on Countries, LLM accuracy drops sharply ($0.79 \rightarrow 0.39$), while the Unsupervised Probe remains stable ($0.80 \rightarrow 0.81$). This suggests that our method is less sensitive to prompt manipulation, which we attribute to its reliance on representation-level geometry rather than autoregressive decoding.

Table 6: Robustness under input perturbations. The Unsupervised Probe is more stable than autoregressive decoding under adversarial system prefixes.

Method	Base	Syntax	Noise	System
LLM output	0.74	0.66	0.69	0.71
Zero-shot probe	0.71	0.66	0.68	0.70
Unsupervised	0.76	0.67	0.68	0.76

E.4 Statistical Significance over Multiple Seeds

We repeat the classification experiments of Table 2 with 20 independent random seeds and report the mean, standard deviation, and 95% confidence intervals. As shown in Table 7, variance is consistently low and confidence intervals are tight (± 0.002 – 0.018) across all tasks and model families, indicating stable performance across runs. The observed improvements of the Unsupervised Probe are not due to stochastic variation but reflect a consistent property of the learned representations.

Table 7: Mean accuracy with standard deviation and 95% confidence intervals over 20 seeds. Each cell shows ACC / Std / CI.

Task	LLaMA-8B	LLaMA-3B	Mistral	GPT2-small
Animals	0.94 / 0.004 / ± 0.002	0.89 / 0.010 / ± 0.005	0.87 / 0.012 / ± 0.006	0.29 / 0.016 / ± 0.008
Countries	0.81 / 0.018 / ± 0.009	0.82 / 0.038 / ± 0.018	0.84 / 0.025 / ± 0.012	0.52 / 0.037 / ± 0.018
Characters	0.60 / 0.013 / ± 0.006	0.47 / 0.016 / ± 0.008	0.54 / 0.010 / ± 0.005	0.26 / 0.017 / ± 0.008
Authors	0.76 / 0.010 / ± 0.005	0.59 / 0.015 / ± 0.007	0.53 / 0.015 / ± 0.007	0.21 / 0.014 / ± 0.007
Languages	0.89 / 0.012 / ± 0.006	0.92 / 0.011 / ± 0.006	0.98 / 0.004 / ± 0.002	0.83 / 0.008 / ± 0.004
Emotions	0.54 / 0.007 / ± 0.003	0.53 / 0.006 / ± 0.003	0.51 / 0.007 / ± 0.003	0.22 / 0.010 / ± 0.005
Fruits	0.70 / 0.014 / ± 0.007	0.65 / 0.017 / ± 0.008	0.64 / 0.012 / ± 0.006	0.39 / 0.017 / ± 0.008
Companies	0.82 / 0.011 / ± 0.005	0.77 / 0.009 / ± 0.004	0.83 / 0.005 / ± 0.003	0.61 / 0.013 / ± 0.006

E.5 Compositional and Multi-hop Task Details

This appendix gives the full task specifications for the compositional and multi-hop experiments summarized in Section 5.6 and Table 5.

Multi-hop chain. Novel words are mapped to categories in-context, and items pass through ownership chains. The model must resolve: person $\rightarrow$ person $\rightarrow$ novel word $\rightarrow$ category. Example ($k=2$): “A blorp is a fruit. [...] Dave bought a blorp. Dave gave his item to Eve. What food group does Eve have?” $\rightarrow$ *blorp*. Prototypes are the novel words (blorp, snark, ...), and mappings change every sample, preventing memorization.

ICL operator tasks. Arbitrary symbols are mapped to operations through few-shot examples. We extract activations at each operator-token position and classify against prototypes (5-way). Three variants:

- *Language translation* (*=Spanish, #=French, ~ =Portuguese, +=German, @=Italian): “apple #tree $\rightarrow$ manzana arbre”. Prototypes: language names. The probe recovers language identity (individual F1 up to 1.0) even when the model only generates correct translations 31% of the time.
- *Vowel append* (*=a, #=e, ~ =i, +=o, @=u): “dog #cat $\rightarrow$ doga cate”. Prototypes: single-vowel tokens.
- *Text operations* (*=capitalize, #=plural, +=repeat, ~ =suffix, @=truncate): “*apple #tree $\rightarrow$ APPLE trees”. Evaluated at the operator-token positions.

Arithmetic composition. Does the token “13” (without context) provide the direction that aligns with the model’s representation when computing “4+9=”? Prototypes are number tokens 3–30. We tested addition and multiplication, restricting to whole-number answers.

SVAMP word problems. Natural-language math requiring multi-step reasoning. We use numbers ≤ 60 as answer classes.

refCOCO. Object description and image. Using LLaVA, we pair each image with its correct referring expression and 9 descriptions from random images (10 classes, random = 10%). We perform inference with a placeholder prompt and each text description separately, extract attention-head outputs at the last-token position, and classify by projection similarity.

E.6 Attention vs. MLP Outputs

To probe the boundary of the linear-interface assumption, we compare attention outputs (linear) against MLP activations (non-linear) on the same classification tasks. Attention consistently outperforms MLP on categorical concepts (Animals, Countries, Emotions). For more complex concepts (Authors, Characters), MLP achieves comparable performance but with less consistency across tasks. We probe MLP outputs only at the last-token position, which may underestimate their performance. These results suggest that MLP outputs, while computed nonlinearly, still produce linearly accessible features in the residual stream—consistent with the fact that MLP outputs are added linearly to the residual stream and must be readable by downstream linear interfaces. The corresponding numbers

are reported as the bottom block of Table 2 in the main text. A more systematic analysis of which feature types admit such behavior (e.g., compositional or abstract reasoning features) is an important direction for future work.

F Supplementary Visualizations

This section provides additional SAE-based analyses supporting the invariant feature sharing and head-level sparsity patterns discussed in the main paper.

Across different datasets and models, we report (i) shared top- k SAE latent dimensions between instance-mean and class-prototype representations within individual heads, and (ii) unsupervised head-level classification performance using SAE latents. Figures are grouped by dataset and model to demonstrate the consistency of these patterns.

F.1 Dataset Explanation

We constructed several datasets to evaluate where semantic attributes are encoded in LLMs and whether semantic features exhibit this linear invariance.

Countries. We analyze the spaces of all the countries in the world grouped by their respective continent.

Languages. To evaluate cross-lingual representations, we employed the XQuAD dataset [Artetxe et al., 2020], a benchmark for cross-lingual question answering. We sampled 200 questions per language, covering English, Spanish, Russian, Hindi, German, and Mandarin Chinese.

Emotions. We used a subset of the Emotion Cause dataset [Ghazi et al., 2015], which includes 1,594 English sentences annotated with seven emotion labels (fear, sadness, anger, happiness, surprise, disgust, and shame). We sampled 600 examples for evaluation.

Cartoon Characters. To probe stylistic and character-specific features, we constructed a dataset centered on fictional characters with distinctive linguistic patterns. For each of six characters—Elmer Fudd, Foghorn Leghorn, Jar Jar Binks, Porky Pig, Scooby-Doo, and Yoda—we collected 50 iconic phrases from publicly available sources.

Literary Authors. We obtained 20 book quotes per author—William Faulkner, Gabriel García Márquez, Ernest Hemingway, Edgar Allan Poe, Virginia Woolf, William Shakespeare, and Mark Twain—from Goodreads.

Animals. This dataset includes animal species names categorized by biological class: mammals, invertebrates, birds, amphibians, reptiles, and fish.

Fruits and Companies. To test polysemy, we compiled a descriptive dataset where tokens like “Apple” belong to both the fruit and company categories, enabling evaluation of how context disambiguates shared representational directions. The company classes were “Apple”, “Google”, “Microsoft”, and “Amazon”, while the fruit classes were “Apple”, “Grape”, “Orange”, and “Banana”..

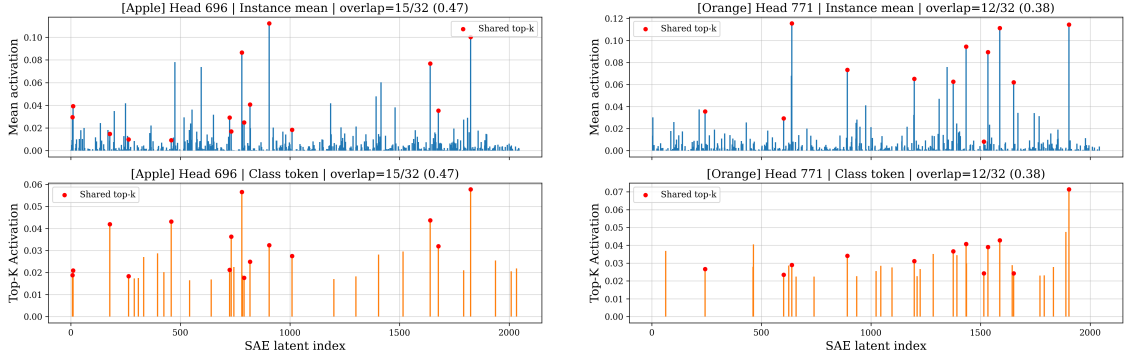


Figure 8: Shared top- k SAE latent activations between instance-mean and class-prototype representations for the Apple and Orange classes in Llama-3-8B, highlighting class-consistent sparse features within individual attention heads.

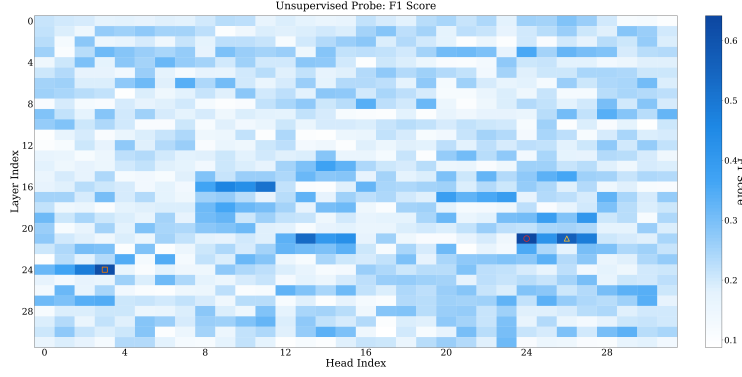


Figure 9: Unsupervised head-level classification performance on the Fruits task using SAE latent representations extracted from Llama-3-8B.

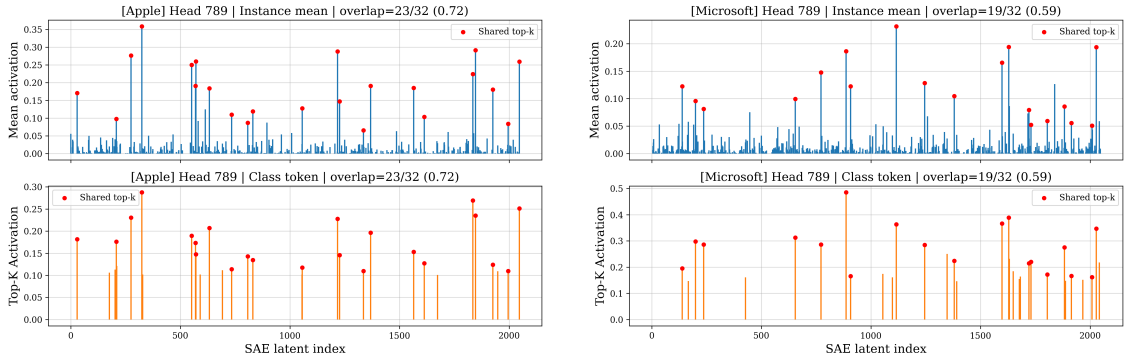


Figure 10: Shared top- k SAE latent activations between instance-mean and class-prototype representations for the Apple and Microsoft classes in Mistral, indicating head-specific sparse features aligned with company semantics.

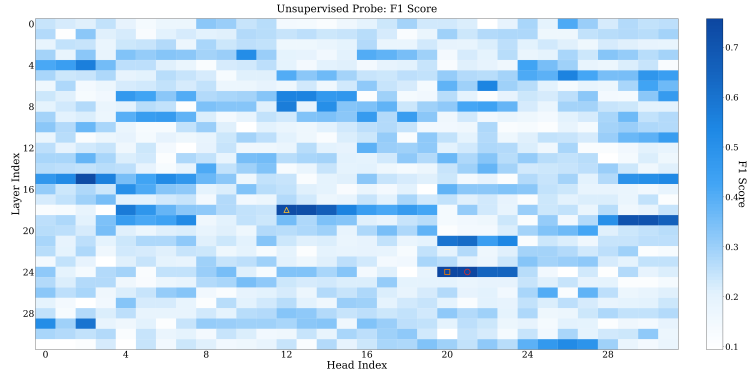


Figure 11: Unsupervised head-level classification performance on the Companies task using SAE latent representations extracted from Mistral.

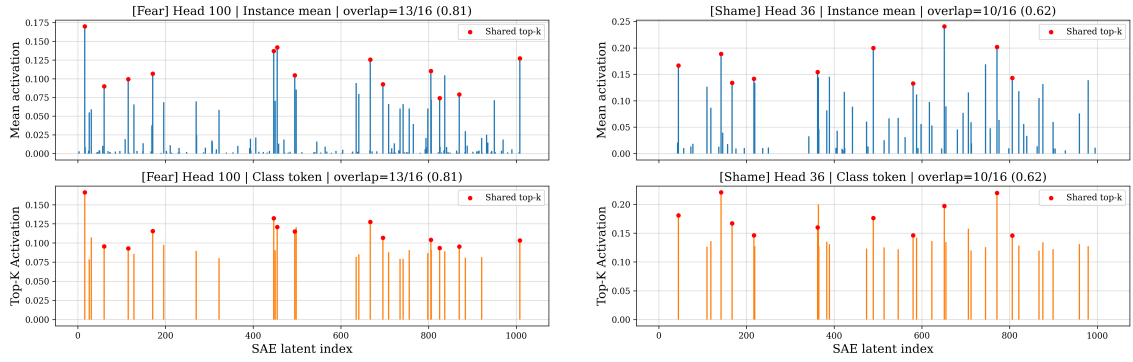


Figure 12: Shared top- k SAE latent activations between instance-mean and class-prototype representations for the Fear and Shame emotion classes in GPT-2.

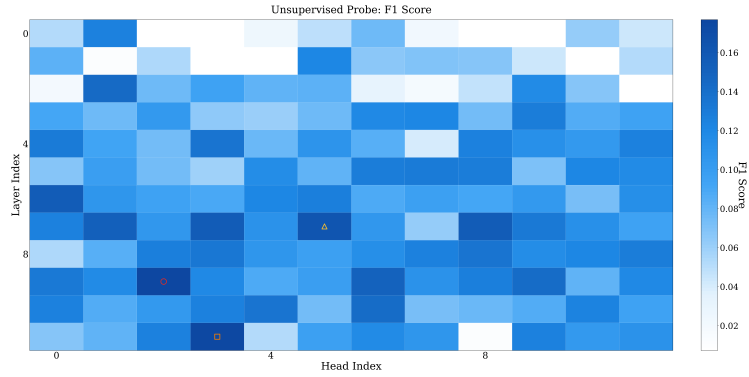


Figure 13: Unsupervised head-level classification performance on the Emotions task using SAE latent representations extracted from GPT-2.

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